

PRACTICE SHEET

- In a manufacture of readymade garments, which average is used to find the most frequent size?
(a) Arithmetic mean (b) Geometric mean
(c) Mode (d) Harmonic mean
- What is the arithmetic mean of the series ${}^nC_0, {}^nC_1, \dots, {}^nC_n$?
(a) $\frac{2^n}{n}$ (b) $\frac{2^n}{(n+1)}$
(c) $\frac{2^{(n+1)}}{n}$ (d) $\frac{2^{(n+1)}}{(n+1)}$
- If from the point of intersection of two gives, a perpendicular is drawn on the x – axis, what does the x – coordinate give?
(a) Arithmetic Mean (b) Mode
(c) Median (d) Geometric Mean
- The definition of Mode fails if:
(a) The maximum frequency is repeated
(b) The maximum frequency is not repeated
(c) The maximum frequency occurs in the middle
(d) The curve drawn with the help of given data is symmetrical
- Which one of the following statement is not correct?
(a) Median divides distributions into two equal subgroups
(b) The third quartile is the same as the 75th percentile
(c) The 5th decile is the same as the 50th percentile
(d) The 50th decile is the same as the 5th percentile
- The mean weight of all the students in a certain class is 60 kg. The mean weight of the boys from the class is 70 kg. While that of the girls is 55 kg. What is the ratio of number of boys to that of girls?
(a) 2 : 1 (b) 1 : 2
(c) 1 : 4 (d) 4 : 1
- If the slopes of the line of regression of Y and X and of X and Y are 30° and 60° respectively, then r (X, Y) is :
(a) -1 (b) 1
(c) $\frac{1}{\sqrt{3}}$ (d) $-\frac{1}{\sqrt{3}}$
- If in a frequency distribution table with 12 classes, the width of each class is 2.5 and the lowest class boundary is 6.1, then what is the upper class boundary of the highest class?
(a) 30.1 (b) 27.6
(c) 30.6 (d) 36.1
- Consider the following statements:
The appropriate number of classes while constructing a frequency distribution should be chosen such that
1. The class – frequency first increases to a peak and then declines.
2. The class – frequency should cluster around the class mid-point.
Which of the statements given is / are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- The populations of four towns A, B, C and D as on 2001 are as follows:

Town	Population
A	6863
B	519
C	12185
D	1785

What is the most appropriate diagram to present the above data?

- (a) Pie diagram (b) Bar chart
(c) Cubic chart (d) Histogram

11. Consider the two series observations A and B as follows:

Series A	1019	1008	1015	1006	1002
Series B	1.9	0.8	1.5	0.6	0.2

If the standard deviation of the Series A is $\sqrt{38}$, then what is the standard deviation of the Series B?

- (a) 3.8 (b) $\sqrt{0.38}$
(c) 0.38 (d) $\sqrt{38}$

12. Let $\bar{X}$ be the mean of n observations $x_1, x_2, \dots, x_n$. If (a – b) is added to each observation, then what is the mean of new set of observations?

- (a) 0 (b) $\bar{X}$
(c) $\bar{X} - (a - b)$ (d) $\bar{X} + (a - b)$

13. What is the values of n for which the numbers 1, 2, 3,, n have variance 2?

- (a) 4 (b) 5
(c) 6 (d) 8

14. The average age of 20 students in a class is 15 yr. If the teacher's age is included, the average increases by one. What is the teacher's age?

- (a) 30 yr (b) 21yr
(c) 42 yr (d) 36 yr

15. For a set of discrete numbers, three measures of central tendency are given below:

1. Arithmetic mean
2. Median
3. Geometric mean

Which of the above measures may not have a meaningful definition?

- (a) 1 only
(b) 2 only
(c) 3 only
(d) All of them are meaningfully defined

16. Consider the following three methods of collecting data

- (1) Collecting data from government office
(2) Collecting data from public libraries
(c) Collecting data by telephonic interview

Select the correct answer using the code given below

- (a) All the three methods give secondary data
(b) 1 and 2 give secondary and 3 gives primary data
(c) 1 and 3 give secondary and 2 gives primary data
(d) 2 and 3 give secondary and 1 gives primary data

17. The arithmetic mean of 4 numbers is 15. The arithmetic mean of another 6 numbers is 12. What is the arithmetic mean of the combined 10 numbers?

- (a) 12.2 (b) 12.8

(c) 13.2

(d) 13.8

18. In a factory, there are 30 men and 20 women employees. If the average salary of men is Rs. 4050 and the average salary of all the employees is Rs. 3550, then what is the average salary of women?

(a) Rs. 3800 (b) Rs. 3300
(c) Rs. 3000 (d) Rs. 2800

19. What is the standard deviation of numbers 7, 9, 11, 13, 15?

(a) 2.2 (b) 2.4
(c) 2.6 (d) 2.8

20. Correlation between two variable is said to be perfect if

(a) One variable increases, the other also increases
(b) One variable increases, the other decreases
(c) One variable increases, the other also increases proportionally
(d) One variable increase, the other decreases proportionally

Directions (for next three): The table below gives an incomplete frequency distribution with two missing frequency f_1 and f_2 .

Value of x	Frequency
0	f_1
1	f_2
2	4
3	4
4	3

The total frequency is 18 and the arithmetic mean of x is 2.

21. What is the value of f_2 ?

(a) 4 (b) 3
(c) 2 (d) 1

22. What is the standard deviation?

(a) $\frac{\sqrt{5}}{2}$ (b) $\frac{\sqrt{5}}{3}$
(c) $\frac{4}{3}$ (d) $\frac{16}{9}$

23. What is the coefficient of variance?

(a) $\frac{200}{3}$ (b) $\frac{50\sqrt{5}}{9}$
(c) $\frac{600}{\sqrt{5}}$ (d) 150

24. The two lines of regression are $8x - 10y = 66$ and $40x - 18y = 214$, and variance of x series is 9. What is the standard deviation of y series?

(a) 3 (b) 4
(c) 6 (d) 9

25. What is the mean deviation of the data 2, 9, 9, 3, 6, 9 and 4?

(a) 2.23 (b) 2.57
(c) 3.23 (d) 3.57

26. The standard deviation of some consecutive integers is found to be 2. Which of the following statements best describes the nature of the consecutive integers?

(a) The integer's deviation of some consecutive integers
(b) The integers are any set of eight consecutive positive integers
(c) The integers are any set of seven consecutive integers

(d) None of the above

27. Consider the following data:

	Factory A	Factory B
Mean wage of workers	540	620
Standard deviation of wages	40.50	31

What is the variability in the wages of the workers in Factory A?

(a) 100% more than the variability in the wages of the workers in Factory B
(b) 50% more than the variability in the wages of the workers in factory B
(c) 50% less than the variability in the wages of the workers in Factory B
(d) 150% more than the variability in the wages of the workers in Factory B

Directions (for next three): The frequency distribution of life of 90 TV tubes whose median life is 17 months is as follows:

Life of TV tubes (in months)	Number of TV tubes
0-5	3
5-10	12
10-15	X
15-20	35
20-25	Y
25-30	4

28. What is the lower limit of the median class?

(a) 10 (b) 15
(c) 20 (d) 25

29. What is the missing frequency y?

(a) 20 (b) 16
(c) 15 (d) 12

30. What is the cumulative frequency of the modal class?

(a) 31 (b) 35
(c) 66 (d) 21

31. The distributions X and Y with total number of observations 36 and 64 and means 4 and 3 respectively are combined. What is the mean of the resulting distribution X + Y?

(a) 3.26 (b) 3.32
(c) 3.36 (d) 3.42

32. A set of n values $x_1, x_2, x_3, \dots, x_n$ has standard deviation σ . What is the standard deviation of n values $x_1 + k, x_2 + k, \dots, x_n + k$?

(a) σ (b) $\sigma + k$
(c) $\sigma - k$ (d) $k\sigma$

33. Consider the following data:

x	5	7	8	4	6
y	2	4	3	2	4

What is the regression equation of y on x?

(a) $y = 0.6 + 0.4x$ (b) $y = 0.7 + 0.3x$
(c) $y = 6 + 5x$ (d) $y = 4 + 9x$

ANSWER KEYS

1.	c	2.	b	3.	c	4.	a	5.	d	6.	b	7.	c	8.	d	9.	b	10.	b
11.	b	12.	d	13.	b	14.	d	15.	d	16.	b	17.	c	18.	d	19.	d	20.	c
21.	a	22.	c	23.	a	24.	b	25.	b	26.	c	27.	c	28.	b	29.	a	30.	c
31.	c	32.	a	33.	a														

Solutions

Sol.1. (c)

Mode is most suitable for this

Sol.2. (b)

Since, expansion contains $(n + 1)$ terms,

Required mean

$$\frac{{}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n}{(n+1)} = \frac{2^n}{n+1}$$

Sol.3. (c)

The x - coordinate of the point of intersection of two ogives, gives median.

Sol.4. (a)

The definition of Mode fails if maximum frequency repeated.

Sol.5. (d)

$$50_{th} \text{ decile} = \frac{50}{10} = 5$$

$$\text{and } 5_{th} \text{ percentile} = \frac{5}{100} \quad 5 \neq \frac{5}{100}$$

Sol.6. (b)

Let there be x number of boys and y number of girls.

Total students = $x + y$

Total weight of the students = $(x + y) 60$

Total weight for boys = $x \times 70$

Total weight for girls = $y \times 55$

Hence, $(x + y) 60 = 70x + 55y$

$60x + 60y = 70x + 55y$, $5y = 10x \Rightarrow y = 2x$

$$\frac{x}{y} = \frac{1}{2} \Rightarrow x : y = 1 : 2$$

Sol.7. (c)

Slope of line of regression of Y and X , is 30° . So

$$b_{yx} = \tan 30^\circ = \frac{1}{\sqrt{3}} \text{ and for } X \text{ and } Y \text{ it is } 60^\circ$$

$$\text{Hence, } \frac{1}{b_{xy}} \tan 60^\circ = \sqrt{3}$$

$$b_{yx} = \frac{1}{\sqrt{3}} \text{ and } b_{xy} = \frac{1}{\sqrt{3}}$$

$$r(x, y) = r^2 = b_{yx} = 1/3$$

$$\text{so, } r = \pm \frac{1}{\sqrt{3}}$$

since, b_{yx} and b_{xy} are both positive $+\frac{1}{\sqrt{3}}$

Sol.8. (d)

Given : lowest class boundary 6.1

Class width = 2.5, Number of classes = 12

$\Rightarrow$ Upper class boundary of the highest class

$$= 6.1 + (2.5 \times 12) = 6.1 + 30 = 36.1$$

Sol.9. (b)

The appropriate number of classes while constructing a frequency distribution should be chosen such that the class frequency should cluster around the class midpoint.

Sol.10. (b)

Bar chart is most appropriate.

Sol.11. (b)

Standard deviation (series B):

$$\sqrt{\frac{1}{5} (1.9^2 + 0.8^2 + 1.5^2 + 0.6^2 + 0.2^2)} = \sqrt{\frac{1}{5} (1.9 + 0.8 + 1.5 + 0.6 + 0.2)^2}$$

$$= \sqrt{\frac{6.9}{5}} - 1 = \sqrt{1.38 - 1}$$

$$= \sqrt{0.38}$$

Sol.12. (d)

Let $\bar{x}$ is the mean of n observation $x_1, x_2, \dots, x_n$.

$$\Rightarrow \bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Now, $(a - b)$ is added to each term

$\therefore$ new mean

$$= \frac{x_1 + (a - b) + x_2 + (a - b) + \dots + x_n + (a - b)}{n}$$

$$= \frac{x_1 + x_2 + \dots + x_n}{n} + \frac{n(a - b)}{n}$$

$$= \bar{x} + (a - b)$$

Sol.13. (b)

$$\text{Mean of the numbers} = \frac{\frac{n(n+1)}{2}}{n}$$

$$= \frac{n+1}{2}$$

$\therefore$ Variance

$$= \frac{\left(1 - \frac{n+1}{2}\right)^2 + \left(2 - \frac{n+1}{2}\right)^2 + \left(3 - \frac{n+1}{2}\right)^2 + \dots}{n}$$

$$2 = \frac{(1^2 + 2^2 + 3^2 + \dots) + n \left(\frac{n+1}{2}\right)^2 - 2 \left(\frac{n+1}{2}\right) [1 + 2 + 3 + \dots]}{n}$$

$$2n = \frac{1}{6} n(n+1)(2n+1) + \frac{n(n+1)^2}{4} - 2 \left(\frac{n+1}{2}\right) \left\{\frac{n(n+1)}{2}\right\}$$

$$2n = n(n+1) \left[\frac{2n+1}{6} + \frac{n+1}{4} - \frac{n+1}{2} \right]$$

$$2 = (n+1) \left[\frac{4n+2-3n-3}{12} \right]$$

$$\Rightarrow 24 = (n+1)(n-1) \Rightarrow n^2 - 1 = 24$$

$$\Rightarrow n^2 = 25 \Rightarrow n = \pm 5$$

Sol.14. (d)

Let the teacher's age is x years

According to question

$$15 + 1 = \frac{20 \times 15 + x}{21}$$

$$\Rightarrow 16 \times 21 = 300 + x$$

$$\Rightarrow x = 336 - 300 = 36 \text{ years}$$

Sol.15. (d)

i. Arithmetic mean

$$= \frac{\text{Sum of all observations}}{\text{Total no. of observation}}$$

ii. Median = The midpoint of the data after being ranked (arranged in ascending order)

iii. Geometric mean = If $x_1, x_2, x_3, \dots, x_n$ are n values of a variate x , none of them being zero, then the geometric mean G is defined as $G = (x_1 x_2 x_3 \dots x_n)^{1/n}$.

Thus, all of them are meaningfully defined.

Sol.16. (b)

Collecting data from government offices is secondary. Collecting data from public libraries is also secondary but collecting data by telephonic interview is primary data.

Sol.17. (c)

Arithmetic mean of

$$10 \text{ numbers} = \frac{4 \times 15 + 6 \times 12}{10} = \frac{60 + 72}{10} = 13.2$$

Sol.18. (d)

Let average salary of women be x .

According to question.

$$\Rightarrow 50 \times 3550 = 30 \times 4050 + 20x$$

$$\Rightarrow 177500 - 121500 = 20x$$

$$\Rightarrow x = 2800$$

Sol.19. (d)

Mean of given numbers

$$= \frac{7+9+11+13+15}{5} = \frac{55}{5} = 11$$

Now,

$$SD = \sqrt{\frac{(7-11)^2 + (9-11)^2 + (11-11)^2 + (13-11)^2 + (15-11)^2}{5}}$$

$$= \frac{\sqrt{16+4+0+4+16}}{5} = \sqrt{8} = 2.8(\text{approx})$$

Sol.20. (c)

Correlation between two variables is said to be perfect, if one variable increases, the other also increases proportionally.

Sol.21. (a)

$$f_2 = 4$$

Sol.22. (c)

$\therefore$ (given)

x	$x - \bar{x}$	$(x - \bar{x})^2$	f	$f(x - \bar{x})^2$
0	-2	4	3	12
1	-1	1	4	4
2	0	0	4	0
3	1	1	4	4
4	2	4	3	12
Total			18	32

$$\text{Now, } SD = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}} = \sqrt{\frac{32}{18}} = \sqrt{\frac{16}{9}} = \frac{4}{3}$$

Sol.23. (a)

Coefficient of variance

$$= \frac{\sigma}{x} \times 100 = \frac{4}{3} \times \frac{1}{2} \times 100 = \frac{200}{3}$$

Sol.24. (b)

The regression coefficient y on x on the line $8x - 10y = 66$, $b_{yx} = 4/5$

The regression coefficient x on y on the line

$$40x - 18y = 214, b_{xy} = \frac{9}{20}$$

$$\Rightarrow r^2 = \frac{36}{100} \Rightarrow r = 0.6$$

$$\Rightarrow b = \frac{r\sigma_y}{\sigma_x}$$

$$\therefore \sigma_y = \frac{\frac{4}{5} \times 3}{0.6} = \frac{12}{3} = 4$$

Sol.25. (b)

$\therefore$ Mean

$$= \frac{2+9+9+3+6+9+4}{7} = \frac{42}{7} = 6$$

$\therefore$ Mean deviation

$$= \frac{|2-6| + |9-6| + |3-6| + |6-6| + |4-6|}{7}$$

$$= \frac{4+9+3+0+2}{7} = \frac{18}{7} = 2.57$$

Sol.26. (c)

Since, the standard deviation of same consecutive integers in 2, then these integers are any set of seven consecutive integers.

Sol.27. (c)

The availability in the wages of the workers in Factory A is 50% less than the variability in the wages of the workers in factory B.

Class	Frequency	cf
0-5	3	3
5-10	12	15
10-15	X	15+x
15-20	35	50+x
20-25	y	50+x+y
25-30	4	54+x+y

$\therefore$

$$\therefore \frac{N}{2} = 45$$

Sol.28. (b)

Lower limit of median class is 15.

Sol.29. (a)

$$\therefore M = I + \frac{\left(\frac{n}{2} - c\right)}{f} \times h$$

$$\therefore 17 = 15 + \frac{[15 - (15+x)]}{35} \times 5$$

$$\Rightarrow 17 = 15 + \frac{30-x}{7}$$

$$\Rightarrow 14 = 30 - x$$

$$x = 16 \quad \therefore x + y = 36$$

$$\therefore y = 36 - 16 = 20$$

Hence, the unknown frequency is 20.

Sol.30. (c)

Cumulative frequency of modal class is: $= 50 + x = 50 + 16 = 66$

Sol.31. (c)

Required mean

$$= \frac{36 \times 4 + 64 \times 3}{36 + 64} = \frac{144 + 192}{100} = \frac{336}{100} = 3.36$$

Sol.32. (a)

Sol.33. (a)

x	y	x^2	y^2	xy
5	2	25	4	10
7	4	49	16	28
8	3	64	9	24
4	2	16	4	8
6	4	36	16	24
$\Sigma x=30$	$\Sigma y=15$	$\Sigma x^2=190$	$\Sigma y^2=49$	$\Sigma xy=94$

$$\text{Here, } \bar{x} = \frac{30}{5} = 6 \text{ and } \bar{y} = \frac{15}{5} = 3$$

$\therefore b_{yx}$

$$= \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2} = \frac{5 \times 94 - 30 \times 15}{5 \times 190 - (30)^2}$$

$$= \frac{470 - 450}{950 - 900} = \frac{20}{50} = \frac{2}{5} = 0.4$$

Here, line of regression is $y - 3 = 0.4(x - 6)$

$$\Rightarrow y = 0.4 \times 3 - 2.4$$

$$\therefore y = 0.4x + 0.6$$

Solution (for next three):

X	F	xf
	f_1	0
1	f_2	f_2
2	4	8
3	4	12
4	3	12
Total	$f_1 + f_2 + 11$	$32 + f_2$

$$\text{Now, } f_1 + f_2 + 11 = 18 \Rightarrow f_1 + f_2 = 7 \quad \dots(i)$$

$$\text{and } \frac{32 + f_2}{18} = 2 \Rightarrow f_2 = 36 - 32 = 4$$

On putting the value of f_2 in Eq. (i), we get $f_1 = 7 - 4 = 3$

NDA PYQ

Directions (for next two): The following table gives the continuous frequency distribution of a continuous variable X.

Class Interval	Frequency
0-10	5
10-20	10
20-30	20
30-40	5
40-50	10

1. What is the median of the above frequency distributions?
 (a) 23 (b) 24
 (c) 25 (d) 26

2. What is the mean of the above frequency distribution?
 (a) 25 (b) 26
 (c) 27 (d) 28

[NDA (I) - 2011]

3.

Class Interval	1-5	6-10	11-15	16-20
Frequency	3	7	6	5

Consider the following statements in respect of the above frequency distribution.

- I. The median is contained in the modal class.
 II. The distribution is bell-shaped

Which of the above statement is/are correct?

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

[NDA (I) - 2011]

4. Consider the following statements with regard to correlation coefficient r between random variables x and y .
 I. $r = +1$ or -1 means there is a linear relation between x and y .

II. $-1 < r \leq 1$ and r^2 is a measure of the linear relationship between the variables.

Which of the statements given above is/are correct?

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

Directions (for next eight): Study the following table and answer the questions that follow:

Year	Male			Female			Total
	Urban	Rural	Total	Urban	Rural	Total	
1995	280	350			310		1350
1996	370		670	180		450	
1997		130	440		190		
1998	400	280		290			
Total				1060	850		

12. What is the total population for the year 1997?
 (a) 810 (b) 830
 (c) 970 (d) 1030

[NDA (II) - 2011]

13. What is the female urban population in the year 1995?
 (a) 390 (b) 410
 (c) 430 (d) 470

[NDA (II) - 2011]

14. What is the urban population in the year 1997?

5. If the values of a set are measured in cm, what will be the unit of variance?
 (a) cm (b) cm^2
 (c) cm^3 (d) No unit

[NDA (I) - 2011]

6. What is the cumulative frequency curve of statistical data commonly called?
 (a) Cartogram (b) Histogram
 (c) Ogive (d) Pictogram

[NDA (I) - 2011]

7. The average daily income of workers of a factory including that the owner is Rs 110. However, if the income of the owner is excluded, the average daily income of remaining 9 worker is Rs 76. What is the daily income of owner.
 (a) Rs 300 (b) Rs 316
 (c) Rs 322 (d) Rs 416

[NDA (I) - 2011]

8. What is the geometric mean of the data 2, 4, 8, 16, 32?
 (a) 2 (b) 4
 (c) 8 (d) 16

[NDA (I) - 2011]

9. A variate X-takes values 2, 3, 4, 2, 5, 4 and 2. What is the mode?
 (a) 2 (b) 3
 (c) 4 (d) 5

[NDA-2011(I)]

10. Which one of the following is the mean of the data given below:

x_i	6	10	14	18	24	28	30
f_i	2	4	7	12	8	4	3

- (a) 17 (b) 18
 (c) 19 (d) 20

[NDA (II) - 2011]

11. If the variance of the data 2, 4, 5, 6, 17 is v_1 , then what is the variance of the data 4, 8, 10, 12, 34?
 (a) v (b) $4v$
 (c) v^2 (d) $2v$

[NDA (II) - 2011]

12. What is the total population for the year 1997?
 (a) 810 (b) 830
 (c) 970 (d) 1030

[NDA (II) - 2011]

13. What is the female urban population in the year 1995?
 (a) 390 (b) 410
 (c) 430 (d) 470

[NDA (II) - 2011]

14. What is the urban population in the year 1997?

- (a) 400 (b) 460
 (c) 490 (d) 510

[NDA (II) - 2011]

15. What is the total population in the year 1998?
 (a) 1000 (b) 1020
 (c) 1040 (d) 1050

[NDA (II) - 2011]

16. What is the difference between the number of females and the number of males of the year 1995?

- (a) 90 (b) 100
(c) 110 (d) 120

[NDA (II) - 2011]

17. In which year the male population minimum?

- (a) 1995 (b) 1996
(c) 1997 (d) 1998

[NDA (II) - 2011]

18. In which year is the female population maximum?

- (a) 1995 (b) 1996
(c) 1997 (d) 1998

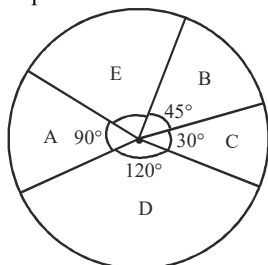
[NDA (II) - 2011]

19. What is the percentage of rural male population (over whole population) in the year 1998?

- (a) $\frac{80}{3}\%$ (b) $\frac{100}{3}\%$
(c) 35% (d) 40%

[NDA (II) - 2011]

Directions (for next four): Study the pie-chart given below and answer the questions that follow:



The pie-chart gives the distribution of funds in a five year plan under the major heads of development expenditures- Agriculture (A), industry (B), Education (C), Employment (D) and Miscellaneous (E). The total allocation is 36000 (in crore of rupees)

20. Which heads is allocated maximum funds?

- (a) Agriculture (b) Industry
(c) Employment (d) Miscellaneous

[NDA (II) - 2011]

21. How much money (in crore) is allocated to Education?

- (a) 3000 (b) 6000
(c) 9000 (d) 10800

[NDA (II) - 2011]

22. How much money (in crore) is allocated to Agriculture and Employment?

- (a) 20000 (b) 21000
(c) 24000 (d) 27000

[NDA (II) - 2011]

23. How much excess money (in crore) is allocated to miscellaneous over Education.

- (a) 3600 (b) 4200
(c) 4500 (d) 4800

[NDA (II) - 2011]

24. What is the geometric mean of 10, 40 and 60?

- (a) $10\sqrt[3]{3}$ (b) $20\sqrt[3]{3}$
(c) $40\sqrt[3]{3}$ (d) $70\sqrt[3]{3}$

[NDA (II) - 2011]

25. What is the median of the distribution 3, 7, 6, 9, 5, 4, 2?

- (a) 5 (b) 6
(c) 7 (d) 8

[NDA (II) - 2011]

26. The mean of 7 observations is 10 and that of 3 observations is 5. What is the mean of all the 10 observations?

- (a) 15 (b) 10

- (c) 8.5 (d) 7.5

[NDA (II) - 2011]

27. Student of three sections of a class, having 30, 30 and 40 students appeared for a test of 100 marks. The arithmetic means of the marks of the three. What is the arithmetic mean of the marks of all the sections are 72.2, 69.0 and 64.1 in that order students of the three sections?

- (a) 66.6 (b) 67.3
(c) 68.0 (d) 70.6

[NDA (II) - 2011]

28. Some measure central tendency for n discrete observations are given below:

- I. Arithmetic mean
II. Geometric mean
III. Harmonic mean
IV. Median

A desirable property of a measure of central tendency is, if every observation is multiplied by c, then the measure of central tendency is also multiplied by c, where $c > 0$.

Which of the above measures satisfy the property.

- (a) I, II and III (b) I, II and IV
(c) III and IV (d) I, II, III and IV

[NDA (II) - 2011]

29. If the arithmetic and geometric means of two numbers are 10, 8 respectively, then one number exceeds the other number by:

- (a) 8 (b) 10
(c) 12 (d) 16

[NDA (II)-2011]

30. What is the mode for the data 20, 20, 20, 21, 21, 21, 21, 22, 22, 22, 22, 22, 22, 23, 23, 23, 23, 24, 24 and 25?

- (a) 7 (b) 21
(c) 22 (d) 25

[NDA (I) - 2012]

31. Consider the following statements:

- I. A continuous random variable can take all values in an interval.
II. A random variable which takes a finite number of values is necessary discrete.
III. Construction of a frequency distribution is based on data which are discrete.

Which of the above statements are correct?

- (a) Both I and II (b) Both II and III
(c) Both I and III (d) All of these

[NDA (I) - 2012]

32. Consider the following statements:

- I. Two independent variables are always correlated.
II. The coefficient of correlation between two variables X and Y is positive when X decrease then Y decrease.

Which of the above statements is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2012]

33. A variate X takes values 2, 9, 3, 7, 5, 4, 3, 2, 10. What is the median?

- (a) 2 (b) 4
(c) 7 (d) 9

[NDA (I) - 2012]

34. What is the arithmetic mean of 16 natural numbers with weights being the numbers itself?

- (a) $17/2$ (b) $33/2$
(c) 11 (d) $187/2$

[NDA (I) - 2012]

35. The geometric mean and harmonic mean of two non-negative observations are 10 and 8 respectively. Then what is the arithmetic mean of the observations equal to?

(a) 4 (b) 9
(c) 12.5 (d) 25

[NDA (I)-2012]

36. The algebraic sum of the deviations of 20 observations measured from 30 is 2. What would be the mean of the observations?

(a) 30 (b) 32
(c) 30.2 (d) 30.1

[NDA (II) - 2012]

37. What is the standard deviation of number 7, 9, 11, 13 and 15?

(a) 2.2 (b) 2.4
(c) 2.6 (d) 2.8

[NDA (II) - 2012]

38. The mean of 10 observations is 5. If 2 is added to each observation and then multiplied by 3, then what will be the new mean?

(a) 5 (b) 7
(c) 15 (d) 21

[NDA (II) - 2012]

39. The median of 27 observations of a variable is 18. Three more observations are made and the values of these observations are 16, 18 and 50. What is the median of these 30 observations?

(a) 18
(b) 19
(c) 25.5
(d) Cannot be determined

[NDA (II) - 2012]

40. Frequency curve may be

(a) Symmetrical (b) Positive skew
(c) Negative skew (d) All of these

[NDA (II) - 2012]

41. What is the mean of n odd natural numbers?

(a) n (b) $\frac{(n+1)}{2}$
(c) $\frac{n(n+1)}{2}$ (d) $n+1$

[NDA (II) - 2012]

42. The arithmetic mean of numbers a, b, c, d, e is M . What is the value of $(a-M) + (b-M) + (c-M) + (d-M) + (e-M)$?

(a) M (b) $a+b+c+d+e$
(c) 0 (d) $5M$

[NDA (II) - 2012]

43. The monthly family expenditure (in%) on n different items are follows:

Food	Rent	Cloth	Transport	Education	Others
38	19	18	-	9	6

if the total monthly expenditure is Rs 9000, then what is the expenditure on transport?

(a) Rs 180 (b) Rs 1000
(c) Rs 900 (d) Rs 360

[NDA (II) - 2012]

44. If the mean of few observation is 40 and standard deviation is 8, then what is the coefficient of variation?

(a) 1% (b) 10%
(c) 20% (d) 30%

[NDA (II) - 2012]

45. Which of the following is measure of dispersion?

(a) Mean
(b) Median

(c) Mode
(d) Standard deviation

[NDA (II) - 2012]

46. Let X and Y be two related variables. The two regression lines are given by $x-y+1=0$ and $2x-y+4=0$. The two regression lines pass through the point:

(a) $(-4, -3)$ (b) $(-6, -5)$
(c) $(3, -2)$ (d) $(-3, -2)$

[NDA (II) - 2012]

47. What is the geometric mean of the sequence 1, 2, 4, 8, ..., 2^n ?

(a) $2^{n/2}$ (b) $2^{(n+1)/2}$
(c) $2^{(n+1)}-1$ (d) $2^{(n-1)}$

[NDA-2012(2)]

48. The marks obtained by 13 students in a test are 10, 3, 10, 12, 9, 7, 9, 6, 7, 10, 8, 6 and 7. The median of this data is:

(a) 7 (b) 8
(c) 9 (d) 10

[NDA (I) - 2013]

49. Consider the following statements:

I. Both variance and standard deviation are measures of variability in the population

II. Standard deviation is the square of the variance.

Which of the above statements is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2013]

50. Variance is always independent of the change of:

(a) Origin but not scale
(b) Only scale
(c) Both origin and scale
(d) None of these

[NDA (I) - 2013]

51. The standard deviation of the observations 5, 5, 5, 5, 5 is:

(a) 0 (b) 5
(c) 20 (d) 25

[NDA (I) - 2013]

52. Consider the following frequency distribution.

Class Interval	Frequency
0-10	14
10-20	X
20-30	27
30-40	Y
40-50	15

If the total of the frequency is 100 and mode is 25, then which one of the following is correct?

(a) $x=2y$ (b) $2x=y$
(c) $x=y$ (d) $x=3y$

[NDA (I) - 2013]

53. The average marks obtained by the students in a class are 43. If the average marks obtained by 25 boys are 40 and the average marks obtained by the girl students are 48, then what is the number of girl students in the class?

(a) 15 (b) 17
(c) 18 (d) 20

[NDA (I) - 2013]

54. If two lines of regression are perpendicular, then the correlation coefficient r is:

(a) 2 (b) $1/2$
(c) 0 (d) None of these

[NDA (I) - 2013]

55. Marks obtained by 7 students in a subject are 30, 55, 75, 90, 50, 60, 39. The number of students securing marks less than the mean marks is

- (a) 7 (b) 6
(c) 5 (d) 4

[NDA (I) - 2013]

56. The harmonic mean H of two numbers is 4 and the arithmetic mean A and geometric mean G satisfy the equation $2A + G^2 = 27$. The two numbers are:

- (a) 6,3 (b) 9,5
(c) 12,7 (d) 3,1

[NDA-2013(1)]

57. The mean of 20 observations is 15. On checking, it was found that two observations were wrongly copied as 3 and 6. If wrong observations are replaced by correct values 8 and 4, then the correct mean is:

- (a) 15 (b) 15.15
(c) 15.35 (d) 16

[NDA (II) - 2013]

58. The arithmetic mean of the square of the first n natural number is:

- (a) $\frac{n(n+1)(2n+1)}{6}$ (b) $\frac{n(n+1)(2n+1)}{2}$
(c) $\frac{(n+1)(2n+1)}{6}$ (d) $\frac{(n+1)(2n+1)}{3}$

[NDA (II) - 2013]

59. Consider the following statements:

I. Both the regression coefficients have same sign.
II. If one of the regression coefficient is greater than unity, the other must be less than unity.

Which of the above statements is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (II) - 2013]

60. Which one of the following measures is determined only after the construction of cumulative frequency distribution?

- (a) Arithmetic mean (b) Mode
(c) Median (d) Geometric mean

[NDA (II) - 2013]

61. Coefficient of correlation is the measure of:

- (a) Central tendency
(b) Dispersion
(c) Both central tendency and dispersion
(d) Neither central tendency nor dispersion.

[NDA (II) - 2013]

62. What is the variance of first 11 natural numbers?

- (a) 10 (b) 11
(c) 12 (d) 13

[NDA (II) - 2013]

63. Consider the following statements in respect of histogram.

I. The algebraic sum of deviations of a set of n values from its arithmetic mean is zero.

II. In the case of frequency distribution, mode is the value of variable which corresponds to maximum frequency.

Which of the above statement (s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither 1 nor 2

[NDA (II) - 2013]

64. Consider the following statements in respect of histogram.

I. Pie diagram is suitable for categorical data.

II. The arc length of a sector of a pie diagram is proportional to the value of the component represented by the sector.

Which of the above statement (s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither 1 nor 2

[NDA (II) - 2013]

65. The variance of 20 observations is 5. If each observations is multiplied by 2, then what is the new variance of the resulting observations?

- (a) 5 (b) 10
(c) 20 (d) 40

[NDA (II) - 2013]

66. The variance of numbers $x_1, x_2, x_3, \dots, x_n$ is V . Consider the following statements.

I. If every x_i is increased by 2, the variance of the new set of numbers is V .

II. If the numbers x_1 is squared, the variance of the new set is V^2 .

Which of the following statements is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

67. What is the mean of the squares of the first 20 natural numbers?

- (a) 151.5 (b) 143.5
(c) 65 (d) 72

[NDA (I) - 2014]

68. p, q, r, s and t are five numbers such that the average of p, q and r is 5 and that of s and t is 10. What is the average of all the five numbers?

- (a) 7.75 (b) 7.5
(c) 7 (d) 5

[NDA (I) - 2014]

Directions (for next three): The number of telephone calls received in 245 successive one minute intervals at an exchange is given below in the following frequency distribution.

Number	0	1	2	3	4	5	6	7
Frequency	14	21	25	43	51	40	39	12

69. What is the mean of the distribution?

- (a) 3.76 (b) 3.84
(c) 3.96 (d) 4.05

[NDA (I) - 2014]

70. What is the median of the distribution?

- (a) 3.5 (b) 4
(c) 4.5 (d) 5

[NDA (I) - 2014]

71. What is the mode of the distribution?

- (a) 3 (b) 4
(c) 5 (d) 6

[NDA (I) - 2014]

72. Let X denote the number of scores which exceed 4 in 18 tosses of a symmetrical die. Consider the following statements.

I. The arithmetic mean of X is 6

II. The standard deviation of X is 2.

Which of the above statements is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

73. The cumulative frequency of the largest observed value must always be:

- (a) Less than the total number of observations
(b) Greater than the total number of observations
(c) Equal to total number of observations
(d) Equal to mid-point of the last class interval

[NDA (I) - 2014]

Directions (for next three): The mean and standard deviation of 100 items are 50, 5 and that of 150 items are 40, 6 respectively.

74. What is the combined mean of all 250 items?
(a) 43 (b) 44
(c) 45 (d) 46
[NDA (I) - 2014]
75. What is the combined standard deviation of all 250 items?
(a) 7.1 (b) 7.3
(c) 7.5 (d) 7.7
[NDA (I) - 2014]
76. What is the variance of all the 250 items?
(a) 50.6 (b) 53.3
(c) 55.6 (d) 59.3
[NDA (I) - 2014]
77. For two variables x and y , two regression coefficients are $b_{yx} = -3/2$ and $b_{xy} = -1/6$. the correlation coefficient between x and y is:
(a) $-1/4$ (b) $1/4$
(c) $-1/2$ (d) $1/2$
[NDA (I) - 2014]
78. What is the mean deviation about the mean for the data 4, 7, 8, 9, 10, 12, 13 and 17?
(a) 2.5 (b) 3
(c) 3.5 (d) 4
[NDA (II) - 2014]
79. The variance of 20 observations is 5. If each observation is multiplied by 2, then what is the new variance of the resulting observation?
(a) 5 (b) 10
(c) 20 (d) 40
[NDA (II) - 2014]
80. Consider the following statements in respect of histogram.
I. The histogram is a suitable representation of a frequency distribution of a continuous variable.
II. The area included under the whole histogram is the total frequency.
Which of the above statement (s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (II) - 2014]
81. If $\bar{x}$ and $\bar{y}$ are the means of two distributions such that $\bar{x} < \bar{y}$ and $\bar{z}$ is mean of the combined distribution, then which one of the following statements is correct?
(a) $\bar{x} < \bar{y} < \bar{z}$ (b) $\bar{x} < \bar{y} < \bar{z}$
(c) $\bar{z} = \frac{\bar{x} + \bar{y}}{2}$ (d) $\bar{x} < \bar{z} < \bar{y}$
[NDA (II) - 2014]
82. The regression lines will be perpendicular to each other, if the coefficient of correlation r is equal to:
(a) Only I (b) 1 or -1
(c) Only -1 (d) 0
[NDA (II) - 2014]
83. Which one of the following measures of central tendency is used in construction of index numbers?
(a) Harmonic mean (b) Geometric mean
(c) Median (d) Mode
[NDA (I) - 2015]
84. Which of the following statements is/are correct in respect of regression coefficients?
I. It measures the degree of linear relationship between two variables.
II. It gives the value by which one variable changes for a unit change in the other variable.
Select the correct answer using the codes given below:
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2015]
85. A set of annual numerical data, comparable over the years, is given for the last 12yr.
Consider the following statements.
I. The data is best represented by a broken line graph, each corner (turning point) representing the data of one year.
II. Such a graph depicts the chronological change and also enables one to make a short-term forecast.
Select the correct answer using the codes given below:
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2015]
86. The mean of five numbers is 30. If one number is excluded, their mean becomes 28. The excluded number is:
(a) 28 (b) 30
(c) 35 (d) 38
[NDA (I) - 2015]
87. The 'less than' ogive curve and the 'more than' ogive curve intersect at:
(a) Median (b) Mode
(c) Arithmetic mean (d) None of these
[NDA (I) - 2015]
88. The correlation coefficient between two variables X and Y is found to be 0.6, All the observations on X and Y are transformed using the transformations $U = 2 - 3X$ and $V = 4Y + 1$. The correlation coefficient between the transformed variables U and V will be
(a) -0.5 (b) +0.5
(c) -0.6 (d) +0.6
[NDA (I) - 2015]
89. The mean and the variance of 10 observations are given to be 4 and 2 respectively. If every observation is multiplied by 2, the mean and the variance of the new series will be respectively.
(a) 8 and 20 (b) 8 and 4
(c) 8 and 8 (d) 80 and 40
[NDA (I) - 2015]
90. The arithmetic mean of 1, 8, 27, 64, ..., upto n terms is given by ?
(a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)^2}{2}$
(c) $\frac{n(n+1)^2}{4}$ (d) $\frac{n^2(n+1)^2}{4}$
[NDA (I) - 2015]
91. The geometric mean of the observations $x_1, x_2, x_3, \dots, x_n$ is G_1 , The geometric mean of the observations $y_1, y_2, y_3, \dots, y_n$ is G_2 . The geometric mean of the observations $\frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, \dots, \frac{x_n}{y_n}$ is
(a) $G_1 G_2$ (b) $\ln(G_1 G_2)$
(c) $\frac{G_1}{G_2}$ (d) $\ln \frac{G_1}{G_2}$
[NDA (I) - 2015]
92. The regression coefficients of a bivariate distribution are -0.64 and -0.36. Then the correlation coefficient of the distribution is:
(a) 0.48 (b) -0.48
(c) 0.50 (d) -0.50
[NDA (II) - 2015]

93. If the total number of observations is 20, $\sum x_i = 1000$ and $\sum x_i^2 = 84000$, then what is the variance of the distribution?
- (a) 1500 (b) 1600
(c) 1700 (d) 1800

[NDA (I) - 2016]

94. What is the mean deviation from the mean of the numbers 10, 9, 21, 16, 24?
- (a) 5.2 (b) 0.5
(c) 0.6 (d) 0.7

[NDA (I) - 2016]

95. The mean of the series $x_1, x_2, \dots, x_n$ is $\bar{X}$. If x_2 is replaced by λ , then what is the new mean?

- (a) $\bar{X} - x_2 + \lambda$ (b) $\frac{\bar{X} - x_2 - \lambda}{n}$
(c) $\frac{\bar{X} - x_2 + \lambda}{n}$ (d) $\frac{n\bar{X} - x_2 + \lambda}{n}$

[NDA (I) - 2016]

96. For the data 3, 5, 1, 6, 5, 9, 5, 2, 8, 6 the mean, median and mode are x, y and z respectively.

Which one of the following is correct?

- (a) $x = y \neq z$ (b) $x \neq y = z$
(c) $x \neq y \neq z$ (d) $x = y = z$

[NDA (I) - 2016]

97. Consider the following statements in respect of a histogram.
1. The total area of the rectangles in a histogram is equal to the total area bounded by the corresponding frequency polygon and the axis.
 2. When class intervals are unequal in a frequency distribution, the area of the rectangle is proportional to the frequency.

Which of the above statements is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2016]

98. If m is the geometric mean of $\left(\frac{y}{z}\right)^{\log(yz)}, \left(\frac{z}{x}\right)^{\log(zx)}$ and $\left(\frac{x}{y}\right)^{\log(xy)}$ then what is the value of m?

- (a) 1 (b) 3
(c) 6 (d) 9

[NDA-2016(I)]

99. Consider the following statements:
1. The mean and median are equal in symmetric distribution.
 2. The range is the difference between the maximum value and the minimum value in the data.
 3. The sum of the area of the rectangles in the histogram is equal to the total area bounded by the frequency polygon and that horizontal axis.

Which of the above statements are correct?

- (a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) All 1, 2 and 3

[NDA (II) - 2016]

100. In a study of two groups, the following results were obtained:

	Group A	Group B
Sample size	20	25
Sample mean	22	23
Sample standard deviation	10	12

Which of the following statements is correct?

- (a) Group A is less variable than group B because group A's standard deviation is smaller.
(b) Group A is less variable than group B because group A's sample size is smaller.
(c) Group A is less variable than group B because group A's sample mean is smaller.
(d) Group A is less variable than group B because group A's coefficient of variation is smaller.

[NDA (II) - 2016]

101. Consider the following statements in respect of class intervals of grouped frequency distribution.

1. Class intervals need not be mutually exclusive.
 2. Class intervals should be exhaustive
 3. Class intervals need not be of equal width
- Which of the above statements are correct?

- (a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2 and 3

[NDA (II) - 2016]

102. A random sample of 20 people is classified in the following table according to their ages:

Age	Frequency
15-25	2
25-35	4
35-45	6
45-55	5
55-65	3

What is the mean age of this group of people?

- (a) 41.0 (b) 41.5
(c) 42.0 (d) 42.5

[NDA (II) - 2016]

103. If the covariance between x and y is 30, variance of x is 25 and variance of y is 144, then what is the correlation coefficient?

- (a) 0.4 (b) 0.5
(c) 0.6 (d) 0.7

[NDA (II) - 2016]

104. The scores of 15 students in an examination were recorded as 10, 5, 8, 16, 18, 20, 8, 10, 16, 20, 18, 11, 16, 14 and 12. After calculating the mean, median and mode, an error is found. One of the values is wrongly written as 16 instead of 18. Which of the following measures of central tendency will change?

- (a) Mean and median
(b) Median and mode
(c) Only mode
(d) Mean and mode

[NDA (II) - 2016]

105. For 10 observations on price (x) and supply (y), the following data was obtained:

$$\sum x = 130, \sum y = 220, \sum x^2 = 2288,$$

$$\sum y^2 = 5506 \text{ and } \sum xy = 3467$$

What is the line of regression of y on x?

- (a) $y = 0.91x + 8.74$ (b) $y = 1.02x + 8.74$
(c) $y = 1.02x - 7.02$ (d) $y = 0.91x - 7.02$

[NDA (II) - 2016]

106. Two variants, x and y, are uncorrelated and have standard deviations σ_x and σ_y respectively. What is the correlation coefficient between $x+y$ and $x-y$?

- (a) $\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$ (b) $\frac{\sigma_x + \sigma_y}{2\sigma_x \sigma_y}$
(c) $\frac{\sigma_x^2 - \sigma_y^2}{\sigma_x^2 + \sigma_y^2}$ (d) $\frac{\sigma_y - \sigma_x}{\sigma_x \sigma_y}$

[NDA (II) - 2016]

107. If $x^{\ln\left(\frac{y}{z}\right)} \cdot y^{\ln(xz)^2} \cdot z^{\frac{\ln x}{y}} = y^{4 \ln y}$ for any $x > 1$, $y > 1$ and $z > 1$, then which one of the following is correct?
- $\ln y$ is the GM of $\ln x$, $\ln x$, $\ln x$ and $\ln z$
 - $\ln y$ is the AM of $\ln x$, $\ln x$, $\ln x$ and $\ln z$
 - $\ln y$ is the HM of $\ln x$, $\ln x$, $\ln x$ and $\ln z$
 - $\ln y$ is the AM of $\ln x$, $\ln x$, $\ln z$ and $\ln z$

[NDA-2016(2)]

108. If two regression lines between height (x) and weight (y) are $4y - 15x + 410 = 0$ and $30x - 2y - 825 = 0$, then what will be the correlation coefficient between height and weight?
- 1/3
 - 1/2
 - 2/3
 - 3/4

[NDA (I) - 2017]

109. In an examination, 40% of candidates got second class. When the data are represented by a pie chart, what is the angle corresponding to second class?
- 40°
 - 90°
 - 144°
 - 320°

[NDA (I) - 2017]

110. Consider the following statements:

Statements:

- Range is not a good measure of dispersion
- Range is highly affected by the existence of extreme values.

Which one of the following is correct in respect of the above statements?

- Both statements 1 and statements 2 are correct and statement 2 is the correct explanation of statement 1.
- Both statements 1 and statement 2 are correct but statement 2 is not the correct explanation of statement 1.
- Statement 1 is correct but statement 2 is not correct.
- Statement 2 is correct but statement 1 is not correct.

[NDA (I) - 2017]

111. If the data are moderately non-symmetrical, then which one of the following empirical relationship is correct?
- $2 \times \text{Standard deviation} = 5 \times \text{Mean deviation}$
 - $5 \times \text{Standard deviation} = 2 \times \text{Mean deviation}$
 - $4 \times \text{Standard deviation} = 5 \times \text{Mean deviation}$
 - $5 \times \text{Standard deviation} = 4 \times \text{Mean deviation}$

[NDA (I) - 2017]

112. Data can be represented in which of the following forms?

- Textual form
- Tabular form
- Graphical form

Select the correct answer using the codes given below:

- 1 and 2
- 2 and 3
- 1 and 3
- 1, 2 and 3

[NDA (I) - 2017]

113. For given statistical data, the graphs for less than ogive and more than ogive are drawn. If the point at which the two curves intersect is P, then abscissa of point P gives the value of which one of the following measures of central tendency?
- Median
 - Mean
 - Mode
 - Geometric mean

[NDA (I) - 2017]

114. If the regression coefficient of x and y and y on x are $-1/2$ and $-1/8$ respectively, then what is the correlation coefficient between x and y?
- $-1/4$
 - $-1/16$
 - $1/16$
 - $1/4$

[NDA (I) - 2017]

115. A sample of 5 observations has mean 32 and median 33. Later it is found that an observation was recorded incorrectly as 40 instead of 35. If we correct the data, then which one of the following is correct?

- The mean and median remain the same.
- The median remains the same but the mean will decrease
- The mean and median both will decrease
- The mean remains the same but median will decrease

[NDA (I) - 2017]

116. The variance of 20 observations is 5. If each observation is multiplied by 3, then what is the new variance of the resulting observations?

- 5
- 10
- 15
- 45

[NDA (I) - 2017]

117. The mean of a group of 100 observations was found to be 20. Later it is found that four observations were incorrect, which were recorded as 21, 21, 18 and 20. What is the mean if the incorrect observations are omitted?

- 18
- 20
- 21
- 22

[NDA (I) - 2017]

118. The mean weight of 150 students in a certain class is 60kg. The mean weight of boys in the class is 70 kg and that of the girls is 55kg. What is the number of boys in the class.

- 50
- 55
- 60
- 100

[NDA (I) - 2017]

119. Consider the following statements:

- Coefficient of variation depends on the unit of measurement of the variable.
- Range is measure of dispersion.
- Mean deviation is least when measured about median.

Which of the above statements are correct?

- 1 and 2
- 2 and 3
- 1 and 3
- 1, 2 and 3

[NDA (II) - 2017]

120. Given that the arithmetic mean and standard deviation of a sample of 15 observations are 24 and 0 respectively. Then which one of the following is the arithmetic mean of the smallest five observations in the data?

- 0
- 8
- 16
- 24

[NDA (II) - 2017]

121. Which one of the following can be considered as appropriate pair of values of regression coefficient of y on x and regression coefficient of x on y?

- (1,1)
- (-1,1)
- $\left(-\frac{1}{2}, 2\right)$
- $\left(\frac{1}{3}, \frac{10}{3}\right)$

[NDA (II) - 2017]

122. It is given that $\bar{X} = 10$, $\bar{Y} = 90$, $\sigma_X = 3$, $\sigma_Y = 12$ and $r_{XY} = 0.8$. Then regression equation of X on Y is:

- $Y = 3.2X + 58$
- $X = 3.2Y + 58$
- $X = -8 + 0.2Y$
- $Y = -8 + 0.2X$

[NDA (II) - 2017]

123. The following table gives the monthly expenditure of two families.

Items	Expenditure (in Rs.)	
	Family A	Family B
Food	3,500	2,700
Clothing	500	800
Rent	1,500	1,000

Education	2,000	1,800
Miscellaneous	2,500	1,800

In constructing a pie diagram to the above data, the radii of the circles are to be chosen by which one of the following ratios?

- (a) 1:1 (b) 10:9
(c) 100:91 (d) 5:4

[NDA (II) - 2017]

124. If a variable takes values 0, 1, 2, 3, ..., n with frequencies 1, C(n,1), C(n,2), C(n,3), ..., C(n,n) respectively, then the arithmetic mean is:

- (a) 2n (b) n+1
(c) n (d) n/2

[NDA (II) - 2017]

125. If x_1 and x_2 are positive quantities, then the condition for the difference between the arithmetic mean and the geometric mean to be greater than 1 is:

- (a) $x_1 + x_2 > 2\sqrt{x_1 x_2}$ (b) $\sqrt{x_1} + \sqrt{x_2} > \sqrt{2}$
(c) $|\sqrt{x_1} - \sqrt{x_2}| > \sqrt{2}$ (d) $x_1 + x_2 < 2(\sqrt{x_1 x_2} + 1)$

[NDA (II) - 2017]

126. Consider the following statements:

1. Variance is unaffected by change of origin and change of scale
2. Coefficient of variance is independent of the unit of observations.

Which of the statements given above is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2017]

127. The coefficient of correlation when coefficients of regression are 0.2 and 1.8 is:

- (a) 0.36 (b) 0.2
(c) 0.6 (d) 0.9

[NDA (II) - 2017]

128. The standard deviation σ of the first N natural numbers can be obtained using which one of the following formulae?

- (a) $\sigma = \frac{N^2 - 1}{12}$ (b) $\sigma = \sqrt{\frac{N^2 - 1}{12}}$
(c) $\sigma = \sqrt{\frac{N - 1}{12}}$ (d) $\sigma = \sqrt{\frac{N^2 - 1}{6N}}$

[NDA (I) - 2018]

129. A train covers the first 5km of its journey at a speed of 30km/hr and the next 15km at a speed of 45km/hr. What is the average speed of the train?

- (a) 35km/hr (b) 37.5km/hr
(c) 39.5km/hr (d) 40 km/hr

[NDA (I) - 2018]

130. The mean and standard deviation of a set of values are 5 and 2 respectively. If 5 is added to each value, then what is the coefficient of variation for the new set of values?

- (a) 10 (b) 20
(c) 40 (d) 70

[NDA (I) - 2018]

131. In a test in Mathematics, 20% of the students obtained "first class". If the data are represented by a Pie-Chart, what is the central angle corresponding to "first class"?

- (a) 20° (b) 36°
(c) 72° (d) 144°

[NDA (I) - 2018]

132. What is the median of a numbers 4.6, 0, 9.3, -4.8, 7.6, 2.3, 12.7, 3.5, 8.2, 6.1, 3.9, 5.2?

- (a) 3.8 (b) 4.9
(c) 5.7 (d) 6.0

[NDA (I) - 2018]

133. Consider the following statements:

- I. The sum of absolute deviations from mean is always zero.
II. The sum of deviations is minimum when taken around median.

Which of the above statement is/are correct?

- (a) I only (b) 2 only
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2018]

134. Consider the following statements:

- I. Mean is independent of change in scale and changes in origin
II. Variance is independent of change in scale but not in origin.

Which of the above statements is/are correct?

- (a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2018]

135. Let $\bar{x}$ be the mean of $x_1, x_2, x_3, \dots, x_n$. If $x_i = a + cy_i$ for some constants a and c, then what will be the mean of $y_1, y_2, y_3, \dots, y_n$?

- (a) $a + c\bar{x}$ (b) $a - \frac{1}{c}\bar{x}$
(c) $\frac{1}{c}\bar{x} - a$ (d) $\frac{\bar{x} - a}{c}$

[NDA (I) - 2018]

136. Consider the following statements:

1. If the correlation coefficient $r_{xy} = 0$, then the two lines of regression are parallel to each other
2. If the correlation coefficient $r_{xy} = +1$, then the two lines of regression are perpendicular to each other

Which of the above statements are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2018]

137. If $4x - 5y + 33 = 0$ and $20x - 9y = 107$ are two lines of regression, then what are the values of $\bar{x}$ and $\bar{y}$ respectively?

- (a) 12 and 18 (b) 18 and 12
(c) 13 and 17 (d) 17 and 13

[NDA (I) - 2018]

138. If the coefficient of correlation between x and y as 0.6, covariance is 27 and variance of y is 25, then what is the variance x?

- (a) 9/5 (b) 81/25
(c) 9 (d) 81

[NDA (I) - 2018]

139. If the ratio of AM to GM of two positive numbers a and b is 5 : 3, then a : b is equal to:

- (a) 3:5 (b) 2:9
(c) 9:1 (d) 5:3

[NDA-2018(I)]

140. In any discrete series (when all values are not same) if x represents mean deviation about mean and y represents standard deviation, then which one of the following is correct?

- (a) $y \geq x$ (b) $y \leq x$
(c) $x = y$ (d) $x < y$

[NDA (II) - 2018]

141. Which one of the following can be obtained from an ogive?

- (a) Mean (b) Median

(c) Geometric

(d) Mode

[NDA (II) - 2018]

142. An analysis of monthly wages paid to the workers in two firms A and B belonging to the same industry the following result:

	Firm A	Firm B
Number of workers	500	600
Average monthly wage	Rs. 1860	Rs. 1750
Variance of distribution of wages	81	100

The average of monthly wages and variance of distribution of wages of all the workers in the firms A and B taken together are:

- (a) Rs.1860, 100 (b) Rs. 1750, 100
(c) Rs. 1800m, 81 (d) None of above

[NDA (II) - 2018]

143. If $x_i > 0$, $y_i > 0$, ($i = 1, 2, 3, \dots, n$) are the values of two variable X and Y with geometric mean P and Q respectively,

then the geometric mean of $\frac{X}{Y}$ is:

- (a) $\frac{P}{Q}$ (b) $\text{Antilog}\left(\frac{P}{Q}\right)$
(c) $n(\log P - \log Q)$ (d) $n(\log P + \log Q)$

[NDA (II) - 2018]

144. The variance of 25 observations is 4. If 2 is added to each observation, then the new variance of the resulting observations is:

- (a) 2 (b) 4
(c) 6 (d) 8

[NDA (II) - 2018]

145. Consider the following statements:

I.If 10 is added to each entry on a list then the average increase by 10.
II.If 10 is added to each entry on a list, then the standard deviation increase by 10
III. If each entry on a list is doubled, then the average doubles.

Which of the above statement are correct?

- (a) I, II and III (b) I and II only
(c) I and 3 only (d) II and III only

[NDA (II) - 2018]

146. The average age of a combined group of men and women is 25 years. If the average age of the group of men is 26 years and the of the group of women is 21 years, then the percentage of men and women in the group is respectively.

- (a) 20, 80 (b) 40, 50
(c) 60, 40 (d) 80, 20

[NDA (II) - 2018]

147. The correlation coefficient computed from a set of 30 observations is 0.8. Then the percentage of variation not explained by linear regression is

- (a) 80% (b) 20%
(c) 64% (d) 36%

[NDA (II) - 2018]

148. If the regression coefficient of Y on X is -6 and the correlation coefficient between X and Y is $-1/2$, then the regression coefficient of X on Y would be

- (a) $1/24$ (b) $-1/24$
(c) $-1/6$ (d) $1/6$

[NDA (II) - 2018]

149. The set of bivariate observations $(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)$ are such that all the values are distinct and all the

observations fall on a straight line with non zero slope. then the possible values of the correlation coefficient between x and y are

- (a) 0 and 1 only (b) 0 and -1 only
(c) 0, 1, -1 (d) -1 and 1 only

[NDA (II) - 2018]

150. If $\sin\beta$ is the harmonic mean of $\sin\alpha$ and $\cos\alpha$, and $\sin\theta$ is the arithmetic mean of $\sin\alpha$ and $\cos\alpha$, then which of the following is/are correct?

$$1. \sqrt{2} \sin\left(\alpha + \frac{\pi}{4}\right) \sin\beta = \sin 2\alpha$$

$$2. \sqrt{2} \sin\theta = \cos\left(\alpha - \frac{\pi}{4}\right)$$

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2018(2)]

151. In which of the following cases would you expect the negative correlation

- (a) The ages of husbands and wives
(b) Shoe size and intelligence
(c) Insurance companies profit and the number of claims they have to pay
(d) Amount of rainfall and yield of crop

[NDA-2018(2)]

152. If two variables X and Y are independent, then what is correlation coefficient between them?

- (a) 1 (b) -1
(c) 0 (d) None

[NDA (I) - 2019]

153. The mean of 100 observations is 50 and the standard deviation is 10. If 5 is subtracted from each observation and then it is divided by 4, then what will be the new mean and new standard deviation respectively?

- (a) 45,5 (b) 11.25,1.25
(c) 11.25, 2.5 (d) 12.5,2.5

[NDA (I) - 2019]

154. Consider the following statements:

1.The algebraic sum of deviations of a set of values from their arithmetic mean is always zero.
2.Arithmetic mean>median>mode for a symmetric distribution.

Which of the above statements are corrects?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2019]

155. Let the correlation coefficient between X and Y be 0.6, random variable Z and W are defined as $Z = X + 5$ and

$$W = \frac{Y}{3}. \text{ What is the correlation coefficient between Z and W?}$$

- (a) 0.1 (b) 0.2
(c) 0.36 (d) 0.6

[NDA (I) - 2019]

156. If all the natural numbers 1 and 20 are multiplied by 3, then what is the variance of the resulting series?

- (a) 99.75 (b) 199.75
(c) 299.25 (d) 399.25

[NDA (I) - 2019]

157. The median of the observations 22, 24, 33, 37, $x + 1$, $x + 3$, 46, 47, 57, 58 in ascending order is 42. What are the values of 5th and 6th observations respectively?

- (a) 42, 45
(b) 41, 43
(c) 43, 46
(d) 40, 40

[NDA (II) - 2019]

158. Arithmetic mean of 10 observations is 60 and sum of squares of deviations from 50 is 5000. What is the standard deviation of the observation?

- (a) 20
(b) 21
(c) 22.36
(d) 24.70

[NDA (II) - 2019]

159. For the variables x and y , the two regression lines are $6x + y = 30$ and $3x + 2y = 25$. What are the values of $\bar{x}$, $\bar{y}$, r respectively?

- (a) 20/3, 35/9, -0.5
(b) 20/3, 35/9, 0.5
(c) 35/9, 20/3, -0.5
(d) 35/9, 20/3, 0.5

[NDA (II) - 2019]

160. The class marks in a frequency table are given to be 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. The class limits of the first five classes are

- (a) 3-7, 7-13, 13-17, 17-23, 23-27
(b) 2.5-7.5, 7.5-12.5, 12.5-17.5, 17.5-22.5, 22.5-27.5
(c) 1.5-8.5, 8.5-11.5, 11.5-18.5, 18.5-21.5, 21.5-28.5
(d) 2-8, 8-12, 12-18, 18-22, 22-28

[NDA (II) - 2019]

161. The mean of 5 observations is 4.4 and variance is 8.24. If three of the five observations are 1, 2 and 6, then what are the other two observations?

- (a) 9, 16
(b) 9, 4
(c) 8, 16
(d) 81, 4

[NDA (II) - 2019]

162. Consider the following discrete frequency distribution:

x	1	2	3	4	5	6	7	8
f	3	15	45	57	50	36	25	9

What is the value of median of the distribution?

- (a) 4
(b) 5
(c) 6
(d) 7

[NDA (II) - 2019]

163. Mean of 100 observations is 50 and standard deviation is 10. If 5 is added to each observation, then what will be the new mean and new standard deviation respectively?

- (a) 50, 10
(b) 50, 15
(c) 55, 10
(d) 55, 15

[NDA (II) - 2019]

164. If the range of a set of observations on a variable X is known to be 25 and if $Y = 40 + 3X$, then what is the range of the set of corresponding observations on Y ?

- (a) 25
(b) 40
(c) 75
(d) 115

[NDA (II) - 2019]

165. If V is the variance and M is the mean of first 15 natural numbers, then what is $V + M^2$ equal to?

- (a) 124/3
(b) 148/3
(c) 248/3
(d) 124/9

[NDA (II) - 2019]

166. A car travels first 60km at a speed of $3v$ km/hr and travels next 60km at $2v$ km/hr. What is the average speed of the car?

- (a) $2.5v$ km/hr
(b) $2.4v$ km/hr
(c) $2.2v$ km/hr
(d) $2.1v$ km/hr

[NDA (II) - 2019]

167. The mean weight of 150 students in a certain class is 60kg. The mean weight of boys is 70kg and that of girls is 55kg.

What are the number of boys and girls respectively in the class?

- (a) 75 and 75
(b) 50 and 100
(c) 70 and 80
(d) 100 and 50

[NDA (II) - 2019]

Direction (for next three)

Read the following information and answer the three items that follow

Marks	Number of students	
	Physics	Mathematics
10-20	8	10
20-30	11	21
30-40	30	38
40-50	26	15
50-60	15	10
60-70	10	6

168. The difference between number of students under Physics and Mathematics is largest for the interval

- (a) 20-30
(b) 30-40
(c) 40-50
(d) 50-60

[NDA - 2020]

169. Consider the following statements :

1. Modal value of the marks in Physics lies in the interval 30 - 40.

2. Median of the marks in Physics is less than that of marks in Mathematics.

Which of the above statements is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

[NDA - 2020]

170. What is the mean of marks in Physics?

- (a) 38.4
(b) 39.4
(c) 40.9
(d) 41.6

[NDA - 2020]

171. What is the standard deviation of the observations $-\sqrt{6}$, $-\sqrt{5}$, $-\sqrt{4}$, -1 , 1 , $\sqrt{4}$, $\sqrt{5}$, $\sqrt{6}$?

- (a) $\sqrt{2}$
(b) 2
(c) $2\sqrt{2}$
(d) 4

[NDA - 2020]

172. If $\sum x_i = 20$, $\sum x_i^2 = 200$ and $n = 10$, for an observed variable x , then what is the coefficient of variation ?

- (a) 80
(b) 100
(c) 150
(d) 200

[NDA - 2020]

173. The arithmetic mean of 100 observations is 40. later it was found that an observation '53' was wrongly read as '83'. What is the correct arithmetic mean?

- (a) 39.8
(b) 39.7
(c) 39.6
(d) 39.5

[NDA - 2020]

174. Let X and Y represent prices (in rupees) of a commodity in Kolkata and Mumbai respectively. It is given that $\bar{X} = 65$, $\bar{Y} = 67$, $\sigma_X = 2.5$, $\sigma_Y = 3.5$ and $r(X, Y) = 0.8$. What is the equation of regression of Y on X ?

- (a) $Y = 0.175X - 5$
(b) $Y = 1.12X - 5.8$
(c) $Y = 1.12X - 5$
(d) $Y = 0.17X + 5.8$

[NDA - 2020]

175. The numbers 4 and 9 have frequencies x and $(x - 1)$ respectively. If their arithmetic mean is 6, then what is the value of x ?

- (a) 2
(b) 3
(c) 4
(d) 5

[NDA - 2020]

176. The sum of n number of observations measured from 2.5 is 50. The sum of deviations of the same set of observations measured from 3.5 is -50 . What is the value of n ?
 (a) 50 (b) 60 (c) 80 (d) 100
[NDA - 2020]
177. A data set of n observations has mean $2M$, while another data set of $2n$ observations has mean M . What is the mean of the combined data sets?
 (a) M (b) $3M/2$ (c) $2M/3$ (d) $4M/3$
[NDA - 2020]
178. If M is the mean of n observations $x_1-k, x_2-k, x_3-k, \dots, x_n-k$, where k is any real number, then what is the mean of $x_1, x_2, x_3, \dots, x_n$?
 (a) M (b) $M+k$ (c) $M-k$ (d) kM
[NDA (I) - 2021]
179. What is the sum of deviation of the variate values 73, 85, 92, 105, 120 from their mean?
 (a) -2 (b) -1 (c) 0 (d) 5
[NDA (I) - 2021]
180. Let x be the HM and y be the GM of two positive numbers m and n . If $5x = 4y$, then which one of the following is correct?
 (a) $5m = 4n$ (b) $2m = n$ (c) $4m = 5n$ (d) $m = 4n$
[NDA (I) - 2021]
181. If the mean of a frequency distribution is 100 and the coefficient of variation is 45%, then what is the value of the variance?
 (a) 2025 (b) 450 (c) 45 (d) 4.5
[NDA (I) - 2021]
182. For which of the following sets of numbers do the mean, median and mode have the same value?
 (a) 12, 12, 12, 12, 24 (b) 6, 18, 18, 18, 30 (c) 6, 6, 12, 30, 36 (d) 6, 6, 6, 12, 30
[NDA (I) - 2021]
183. The mean of 12 observations is 75. If two observations are discarded, then the mean of the remaining observations is 65. What is the mean of the discarded observations?
 (a) 250 (b) 125 (c) 120 (d) Cannot be determined
[NDA (I) - 2021]
184. The geometric mean of a set of observations is computed as 10. The geometric mean obtained when each observation x_i is replaced by $3x_i^4$ is
 (a) 810 (b) 900 (c) 30000 (d) 81000
[NDA (I) - 2021]
185. The average of a set of 15 observations is recorded, but later it is found that for one observation, the digit in the tens place was wrongly recorded as 8 instead of 3. After correcting the observation, the average is
 (a) Reduced by $1/3$ (b) Increased by $10/3$ (c) Reduced by $10/3$ (d) Reduced by 50
[NDA (I) - 2021]
186. If the mode of the scores 10, 12, 13, 15, 15, 13, 12, 10, x is 15, then what is the value of x ?
 (a) 10 (b) 12 (c) 13 (d) 15
[NDA (I) - 2021]
187. Consider the following measures of central tendency for a set of N numbers:
 1. Arithmetic mean
 2. Geometric mean
 Which of the above uses/use all the data?
 (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (I) - 2021]
188. The numbers of science, Arts and Commerce graduates working in a company are 30, 70 and 50 respectively. If these figures are represented by a pie chart, then what is the angle corresponding to Science graduates?
 (a) 36° (b) 72° (c) 120° (d) 168°
[NDA (I) - 2021]
189. For histogram based on a frequency distribution with unequal class intervals, the frequency of a class should be proportional to
 (a) The height of the rectangle (b) The area of the rectangle (c) The width of the rectangle (d) The perimeter of the rectangle
[NDA (I) - 2021]
190. The coefficient of correlation is independent of
 (a) Change of scale only (b) Change of origin only (c) Both change of scale and change of origin (d) Neither change of scale nor change of origin
[NDA (I) - 2021]
191. The following table gives the frequency distribution of number of peas per pea pod of 198 pods:
- | Number of peas | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|----------------|---|----|----|----|----|---|---|
| Frequency | 4 | 33 | 76 | 50 | 26 | 8 | 1 |
- What is the median of this distribution?
 (a) 3 (b) 4 (c) 5 (d) 6
[NDA (I) - 2021]
192. When the measure of central tendency is available in the form of mean, which one of the following is the most reliable and accurate measure of variability?
 (a) Range (b) Mean deviation (c) Standard deviation (d) Quartile deviation
[NDA (II) - 2021]
- Direction:** Consider the following for the next two (02) items that follow:
 Two regression lines are given as $3x - 4y + 8 = 0$ and $4x - 3y - 1 = 0$
193. Consider the following statements:
 1. The regression line of y on x is $y = 3/4x + 2$
 2. The regression line of x on y is $x = 3/4y + 1/4$
 Which of the above statements is/are correct?
 (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (II) - 2021]
194. Consider the following statements:
 1. The coefficient of correlations r is $\frac{3}{4}$
 2. The means of x and y are 3 and 4 respectively.
 Which of the above statement is/are correct?

- (a) 1 only
(c) Both 1 and 2

- (b) 2 only
(d) Neither 1 and 2

[NDA (II) - 2021]

Direction : Consider the following for the next two (02) items that follow:

The marks obtained by 60 students in a certain subject out of 75 are given below:

Marks	Number of students
15-20	4
20-25	5
25-30	11
30-35	6
35-40	5
40-45	8
45-50	9
50-55	6
55-60	4
60-65	2

195. What is the median?

- (a) 35
(c) 39
- (b) 38
(d) 40

[NDA (II) - 2021]

196. What is the mode?

- (a) 27.27
(c) 27.93
- (b) 27.73
(d) 28.27

[NDA (II) - 2021]

197. What is the mean of natural numbers contained in the interval [15, 64]?

- (a) 36.8
(c) 39.5
- (b) 38.3
(d) 40.3

[NDA (II) - 2021]

198. For the set of numbers $x, x, x+2, x+3, x+10$ where x is a natural number, which of the following is/are correct?

1. Mean > Mode
2. Median > Mean

Select the correct answer using the code given below:

- (a) 1 only
(c) Both 1 and 2
- (b) 2 only
(d) Neither 1 nor 2

[NDA (II) - 2021]

199. The mean of 10 observation is 5.5, if each observation is multiplied by 4 and subtracted from 44, then what is the new mean?

- (a) 20
(c) 34
- (b) 22
(d) 44

[NDA (II) - 2021]

200. If g is the geometric mean of 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, then which one of the following is correct?

- (a) $8 < g < 16$
(c) $32 < g < 64$
- (b) $16 < g < 32$
(d) $g > 64$

[NDA (II) - 2021]

201. If the harmonic mean of 60 and x is 48, then what is the value of x ?

- (a) 32
(c) 40
- (b) 36
(d) 44

[NDA (II) - 2021]

202. What is the mean deviation of first 10 even natural numbers?

- (a) 5
(c) 10
- (b) 5.5
(d) 10.5

[NDA (II) - 2021]

203. If $\sum_{i=1}^{10} x_i = 110$ and $\sum_{i=1}^{10} x_i^2 = 1540$ then what is the variance?

- (a) 22
(b) 33

- (c) 44
(d) 55

[NDA (II) - 2021]

204. The frequency curve (assuming unimodal) corresponding to the data obtained in an experiment is skewed to the left. What conclusion can be drawn from the curve?

- (a) Mean > Median > Mode
(b) Mean > Mode > Median
(c) Median > Mean > Mode
(d) Mode > Median > Mean

[NDA (I) - 2022]

205. The variance of five positive observations is 3.6. If four of the observations are 2, 2, 4, 5 then what is the remaining observation?

- (a) 4
(c) 7
- (b) 5
(d) 9

[NDA (I) - 2022]

206. What is the arithmetic mean of 50 terms of an AP with first term 4 and common difference 4?

- (a) 50
(c) 100
- (b) 51
(d) 102

[NDA (I) - 2022]

207. What is the coefficient of mean deviation of 21, 34, 23, 39, 26, 37, 40, 20, 33, 27 (taken from mean)?

- (a) 0.11
(c) 0.33
- (b) 0.22
(d) 0.44

[NDA (I) - 2022]

Consider the following for the next three (03) items that follow:

The algebraic sum of the deviations of a set of values $x_1, x_2, x_3, \dots, x_n$ measured from 100 is -20 and the algebraic sum of the deviations of the same set of values measured from 92 is 140.

208. What is the mean of the values?

- (a) 91
(c) 98
- (b) 96
(d) 99

[NDA (I) - 2022]

209. What is the algebraic sum of the deviations of the same set of values measured from 99?

- (a) 0
(c) 20
- (b) 10
(d) 40

[NDA (I) - 2022]

210. If the algebraic sum of the deviations of the same set of values measured from y is 180, then what is the value of y ?

- (a) 80
(c) 90
- (b) 85
(d) 95

[NDA (I) - 2022]

Consider the following data for the next three (03) items that follows:

The marks obtained by 51 students in a class are in AP with its first term 4 and common difference 3.

211. What is the mean of the marks?

- (a) 67
(c) 75
- (b) 71
(d) 79

[NDA (I) - 2022]

212. What is the median of the marks?

- (a) 79.5
(c) 78.5
- (b) 79
(d) 77

[NDA (I) - 2022]

213. What is the sum of the deviations measured from the median?

- (a) -1
(c) 0
- (b) 0
(d) 2

[NDA (I) - 2022]

Consider the following frequency distribution for the next three (03) items that follow:

Class	0-20	20-40	40-60	60-80	80-100
Frequency	17	p+q	32	p-3q	19

The total frequency is 120. The mean is 50.

214. What is the value of p?
(a) 25 (b) 26
(c) 27 (d) 28

[NDA (I) - 2022]

215. What is the value of q?
(a) 1 (b) 2
(c) 3 (d) 4

[NDA (I) - 2022]

216. If the frequency of each class is doubled, then what would be the mean?
(a) 25 (b) 50
(c) 75 (d) 100

[NDA (I) - 2022]

217. Let x be the mean of squares of first n natural numbers and y be the square of mean of first n natural numbers.

If $\frac{x}{y} = \frac{55}{42}$, then what is the value of n ?

- (a) 24 (b) 25
(c) 27 (d) 30

[NDA 2022 (II)]

218. The coefficient of correlation between ages of husband and wife at the time of marriage for a given set of 100 couples was noted to be 0.7. Assume that all these couples survive to celebrate the silver jubilee of their marriage. The coefficient of correlation at that point of time will be:

- (a) 1 (b) 0.9
(c) 0.7 (d) 0.3

[NDA 2022 (II)]

Consider the following for the next two (02) items that follows:

The mean and standard deviation (SD) of marks obtained by 50 students of a class in 4 subjects are given below:

Subject	Mathematics	Physics	Chemistry	Biology
Mean Marks	40	28	38	36
SD	15	12	14	16

219. Which one of the following subjects shows highest variability of marks?
(a) Mathematics (b) Physics
(c) Chemistry (d) Biology

[NDA 2022 (II)]

220. What is the coefficient of variation of marks in Mathematics?
(a) 37.5% (b) 38.0%
(c) 38.5% (d) 39.0%

[NDA 2022 (II)]

Consider the following for the next three (03) items that follows:

Consider the following grouped frequency distribution:

Class	Frequency
0-10	1
10-20	2
20-30	4
30-40	6
40-50	4
50-60	3

221. What is the median of the distribution?
(a) 34 (b) 34.5
(c) 35 (d) 35.5

[NDA 2022 (II)]

222. What is mean deviation about the median?
(a) 11.4 (b) 11.1
(c) 10.8 (d) 10.5

[NDA 2022 (II)]

223. What is the mean deviation about the mean?
(a) 10.15 (b) 10.65
(c) 11.15 (d) 11.65

[NDA 2022 (II)]

224. Consider the following frequency distribution:

x	1	2	3	5
f	4	6	9	7

What is the value of median of the distribution?

- (a) 1 (b) 2
(c) 3 (d) 3.5

[NDA - 2023 (1)]

225. For data -1, 1, 4, 3, 8, 12, 17, 19, 9, 11; if M is the median of first 5 observations and N is the median of last five observations, then what is the value of $4M - N$?

- (a) 7 (b) 4
(c) 1 (d) 0

[NDA - 2023 (1)]

226. Let P , Q , R represent mean, median and mode. If for some distribution $5P = 4Q = R/2$, then what is $\frac{P+Q}{2P+0.7R}$ equal to?

- (a) 1/12 (b) 1/7
(c) 2/9 (d) 1/4

[NDA - 2023 (1)]

227. If G is the geometric mean of numbers 1, 2, 2^2 , 2^3 , ..., 2^{n-1} , then what is the value of $1 + 2\log_2 G$?

- (a) 1 (b) 4
(c) $n-1$ (d) n

[NDA - 2023 (1)]

228. If H is the harmonic mean of numbers 1, 2, 2^2 , 2^3 , ..., 2^{n-1} , then what is n/H equal to?

- (a) $2 - \frac{1}{2^{n+1}}$ (b) $2 - \frac{1}{2^{n-1}}$
(c) $2 + \frac{1}{2^{n-1}}$ (d) $2 - \frac{1}{2^n}$

[NDA - 2023 (1)]

229. Let P be the median, Q be the mean and R be the mode of observations $x_1, x_2, x_3, \dots, x_n$. Let $S = \sum_{i=1}^n (2x_i - a)^2 \cdot S$ takes minimum value, when a is equal to?

- (a) P (b) $Q/2$
(c) $2Q$ (d) R

[NDA - 2023 (1)]

230. The mean and variance of five observations are 14 and 13.2 respectively. Three of the five observations are 11, 16 and 20. What are the other two observations?

- (a) 8 and 15 (b) 9 and 14
(c) 10 and 13 (d) 11 and 12

[NDA - 2023 (1)]

231. If two lines of regression are $x + 4y + 1 = 0$ and $4x + 9y + 7 = 0$, then what is the value of x when $y = -3$?

- (a) -13 (b) -5
(c) 5 (d) 7

[NDA - 2023 (1)]

232. The central angles p , q , r and s (in degrees) of four sectors in a Pie Chart satisfy the relation $9p = 3q = 2r = 6s$. What is the value of $4p - q$?
- (a) 12 (b) 24
(c) 30 (d) 36
- [NDA – 2023 (1)]
233. The observations 4, 1, 4, 3, 6, 2, 1, 3, 4, 5, 1, 6 are outputs of 12 dices thrown simultaneously. If m and M are means of lowest 8 observations and highest 4 observations respectively, then what is $(2m + M)$ equal to?
- (a) 10 (b) 12
(c) 17 (d) 21
- [NDA – 2023 (1)]
234. A bivariate data set contains only two points $(-1, 1)$ and $(3, 2)$. What will be the line of regression of y on x ?
- (a) $x - 4y + 5 = 0$ (b) $3x + 2y - 1 = 0$
(c) $x + 4y + 1 = 0$ (d) $5x - 4y + 1 = 0$
- [NDA – 2023 (1)]
235. A die is thrown 10 times and obtained the following outputs: 1, 2, 1, 1, 2, 1, 4, 6, 5, 4
What will be the mode of data so obtained?
- (a) 6 (b) 4
(c) 2 (d) 1
- [NDA – 2023 (1)]
- Consider the following for the next two (02) items that follow:
- $$(3 + 2\sqrt{2})x^2 - (4 + 2\sqrt{3})x + (8 + 4\sqrt{3}) = 0$$
236. What is the HM of the roots of the equation.
- (a) 2 (b) 4
(c) $2\sqrt{2}$ (d) $2\sqrt{3}$
- [NDA – 2023 (1)]
237. What is the GM of the roots of the equation?
- (a) $\sqrt{2}(\sqrt{6} - \sqrt{3} + \sqrt{2} - 1)$ (b) $\sqrt{2}(\sqrt{6} + \sqrt{3} - \sqrt{2} - 1)$
(c) $(\sqrt{6} - \sqrt{3} + \sqrt{2} - 1)$ (d) $(\sqrt{6} + \sqrt{3} + \sqrt{2} - 1)$
- [NDA – 2023 (1)]
238. The sum of deviations of a n numbers from 10 and 20 are p and q respectively. If $(p - q)^2 = 10000$, then what is the value of n ?
- (a) 10 (b) 20
(c) 50 (d) 100
- [NDA-2023 (2)]
239. If $\bar{X} = 20$ is the mean of 10 observations $x_1, x_2, \dots, x_{10}$; then what is the value of $\sum_{i=1}^{10} \left(\frac{3x_i - 4}{5} \right)$?
- (a) 0 (b) 12
(c) 112 (d) 1012
- [NDA-2023 (2)]
240. If the mean and the sum of squares of 10 observations are 40 and 16160 respectively, then what is the standard deviation?
- (a) 16 (b) 6
(c) 5 (d) 4
- [NDA-2023 (2)]
241. What is the mean of the numbers 1, 2, 3, ..., 10 with frequencies ${}^9C_0, {}^9C_1, {}^9C_2, \dots, {}^9C_9$ respectively?
- (a) 1.1×2^8 (b) 1.2×7^4
(c) 5.5 (d) 0.55
- [NDA-2023 (2)]
242. If r is the coefficient of correlation between x and y , then what is the correlation coefficient between $(3x+4)$ and $(-3y+3)$?
- (a) $-r$ (b) r
(c) $\sqrt{3}r$ (d) $-\sqrt{3}r$
- [NDA-2023 (2)]
243. If H is the Harmonic Mean of three numbers ${}^{10}C_4, {}^{10}C_5$, and ${}^{10}C_6$, then what is the value of $270/H$?
- (a) 1 (b) $14/17$
(c) $17/14$ (d) $1/31$
- [NDA-2023 (2)]
244. From data $(-4, 1), (-1, 2), (2, 7)$ and $(3, 1)$ the regression line of y on x is obtained as $y = a + bx$, then what is the value of $2a + 15b$?
- (a) 6 (b) 11
(c) 17 (d) 21
- [NDA-2024 (1)]
245. Let $x + 2y + 1 = 0$ and $2x + 3y + 4 = 0$ are two lines of regression computed from some bivariate data. If θ is the acute angle between them, then what is the value of $488 \tan 3\theta$?
- (a) 191 (b) 161
(c) 131 (d) 121
- [NDA-2024 (1)]
246. An edible oil is sold at the rates 150, 200, 250, 300 rupees per litre in four consecutive years. Assuming that an equal amount of money is spent on oil by a family in every year during these years, what is the average price of oil in rupees (approximately) per litre?
- (a) 210 (b) 220
(c) 230 (d) 240
- [NDA-2024 (1)]
- Consider the following for the next two (02) items that follow:
- | | | | | | |
|-------|---|----------|----------|------|--------------|
| x_i | 1 | 2 | 3 | ... | n |
| f_i | 1 | 2^{-1} | 2^{-2} | | $2^{-(n-1)}$ |
247. What is $\sum_{i=1}^n x_i f_i$ equal to?
- (a) $\frac{2^{n+1} - n + 2}{2^{n-1}}$ (b) $\frac{2^{n+1} - n - 2}{2^{n-1}}$
(c) $\frac{2^{n+1} + n + 2}{2^{n-1}}$ (d) $\frac{2^{n+1} - n - 2}{2^n}$
- [NDA-2024 (1)]
248. What is the mean of the distribution?
- (a) $\frac{2^{n+1} - n + 2}{2^n - 1}$ (b) $\frac{2^{n+1} - n - 2}{2^{n-1}}$
(c) $\frac{2^{n+1} - n - 2}{2^n - 1}$ (d) $\frac{2^{n+1} - n + 2}{2^n}$
- [NDA-2024 (1)]
- Consider the following for the next two (02) items that follow:
- The marks obtained by 10 students in a Statistics test are 24, 47, 18, 32, 19, 15, 21, 35, 50 and 41.
249. What is the mean deviation of the largest five observations?
- (a) 4.8 (b) 5.5
(c) 6 (d) 7.5
- [NDA-2024 (1)]
250. What is the variance of the largest five observations?
- (a) 14.6 (b) 21.8
(c) 25.2 (d) 46.8
- [NDA-2024 (1)]
251. Let $\sum_{i=1}^9 x_i^2 = 855$. If M is the mean and σ is the standard deviation if $x_1, x_2, x_3, \dots, x_9$, then what is the value of $M^2 + \sigma^2$?

- (a) 100 (b) 95
(c) 90 (d) 85

[NDA-2024 (2)]

252. The mean of the series $x_1, x_2, x_3, \dots, x_n$ is $\bar{x}$. If x_n is replaced by k , then what is the new mean

- (a) $\bar{x} - x_n + k$ (b) $\frac{n\bar{x} - \bar{x} + k}{n}$
(c) $\frac{\bar{x} - x_n - k}{n}$ (d) $\frac{n\bar{x} - x_n + k}{n}$

[NDA-2024 (2)]

253. let $x - 3y + 4 = 0$ and $2x - 7y + 8 = 0$ be two lines of regression computed from some bivariate data. If b_{yx} and b_{xy} are regression coefficients of lines of regression of y on x and x on y respectively, then what is the value of $b_{xy} + 7b_{yx}$?

- (a) -2 (b) 1
(c) 2 (d) 5

[NDA-2024 (2)]

254. The mean of n observations $1, 4, 9, 16, \dots, n^2$ is 130. What is the value of n ?

- (a) 18 (b) 19
(c) 20 (d) 21

[NDA-2024 (2)]

255. What is the mean deviation of the first 10 natural numbers?

- (a) 2 (b) 2.5
(c) 3 (d) 3.5

[NDA-2024 (2)]

Consider the following for the four (04) items that follow:
The frequency distribution of height of students of a class is given below:

Height (in cm)	Number of Students
160-162	12
162-164	15
164-166	24
166-168	13

256. What is the total number of students whose height is less than or equal to 165 cm?

- (a) 15 (b) 39
(c) 51 (d) None of the above

[NDA-2025 (1)]

257. What is the median height of the class?

- (a) 162.41 cm (b) 163.41 cm
(c) 164.41 cm (d) 165.41 cm

[NDA-2025 (1)]

258. The height which occurs most frequently in the class is

- (a) 163.5 cm (b) 163.9 cm
(c) 164.5 cm (d) 164.9 cm

[NDA-2025 (1)]

259. The most appropriate graphical representation of the given frequency distribution is

- (a) bar chart (b) percentage bar chart
(c) histogram (d) pie chart

[NDA-2025 (1)]

Consider the following for the two (02) items that follow:

The sum and the sum of squares of the observations corresponding to length X (in cm) and weight Y (in gm) of 50 tropical tubers are given as: $\sum X = 200$, $\sum Y = 250$, $\sum X^2 = 900$ and $\sum Y^2 = 1400$.

260. Which one of the following is correct?

- (a) Variance (X) > Variance (Y)
(b) Variance (X) < Variance (Y)
(c) Variance (X) = Variance (Y)
(d) Cannot be determined from the given data

[NDA-2025 (1)]

261. Which one of the following statements is correct?

- (a) Coefficient of variation of X is strictly more than coefficient of variation of Y .
(b) Coefficient of variation of X is strictly less than coefficient of variation of Y .
(c) Coefficient of variation of X is same as coefficient of variation of Y .
(d) Coefficient of variation cannot be determined from the given data.

[NDA-2025 (1)]

262. The arithmetic mean of 100 observations is 50. If 5 is subtracted from each observation and then divided by 20, then what is the new arithmetic mean?

- (a) 2.25 (b) 3.5
(c) 4.25 (d) 5.5

[NDA-2025 (1)]

263. The standard deviation of 100 observations is 10. If 5 is added to each observation and then divided by 20, then what will be the new standard deviation?

- (a) 0.25 (b) 0.5
(c) 0.75 (d) 1.00

[NDA-2025 (1)]

264. What is the arithmetic mean of $8^2, 9^2, 10^2, \dots, 15^2$?

- (a) 133.5 (b) 135.5
(c) 137.5 (d) 139.5

[NDA-2025 (1)]

265. What is the harmonic mean of the numbers $C(10,3)$, $C(10,4)$, $C(10,5)$, $C(10,6)$ and $C(10,7)$?

- (a) 3150/19 (b) 4000/19
(c) 252 (d) 225

[NDA-2025 (2)]

266. The standard deviation of 100 observations is 10. If 20 is added to each observation, then what will be the new standard deviation?

- (a) 10 (b) 15
(c) 20 (d) 25

[NDA-2025 (2)]

267. The frequency distribution of the marks obtained by students in a Science examination is given below:

Marks	Number of students
5-15	20
15-25	30
25-35	30
35-45	20

What is the arithmetic mean?

- (a) 20 (b) 25
(c) 30 (d) 35

[NDA-2025 (2)]

268. Consider the following statements:

- I. Mean and variance have the same unit of measurement.
II. Mean deviation and standard deviation have the same unit of measurement.

Which of the statements given above is/are correct?

- (a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA-2025 (2)]

ANSWER KEY

1.	c	2.	b	3.	b	4.	c	5.	b	6.	c	7.	d	8.	c	9.	a	10.	c
11.	b	12.	a	13.	b	14.	c	15.	d	16.	a	17.	c	18.	a	19.	a	20.	c
21.	a	22.	b	23.	c	24.	b	25.	a	26.	c	27.	c	28.	d	29.	c	30.	c
31.	a	32.	b	33.	b	34.	c	35.	c	36.	d	37.	d	38.	d	39.	a	40.	d
41.	a	42.	c	43.	c	44.	c	45.	d	46.	d	47.	a	48.	b	49.	d	50.	a
51.	a	52.	c	53.	a	54.	c	55.	d	56.	a	57.	b	58.	c	59.	c	60.	c
61.	d	62.	a	63.	c	64.	c	65.	c	66.	a	67.	b	68.	c	69.	a	70.	b
71.	b	72.	c	73.	c	74.	b	75.	c	76.	c	77.	c	78.	b	79.	c	80.	a
81.	d	82.	d	83.	a	84.	c	85.	b	86.	d	87.	a	88.	c	89.	c	90.	c
91.	c	92.	b	93.	c	94.	a	95.	d	96.	d	97.	a	98.	a	99.	d	100.	d
101.	a	102.	b	103.	b	104.	d	105.	b	106.	c	107.	b	108.	b	109.	c	110.	a
111.	c	112.	d	113.	a	114.	a	115.	b	116.	d	117.	b	118.	a	119.	b	120.	d
121.	a	122.	c	123.	b	124.	d	125.	c	126.	b	127.	c	128.	b	129.	d	130.	b
131.	c	132.	b	133.	c	134.	d	135.	d	136.	d	137.	c	138.	d	139.	c	140.	d
141.	b	142.	d	143.	a	144.	b	145.	c	146.	d	147.	d	148.	b	149.	d	150.	c
151.	c	152.	c	153.	c	154.	a	155.	d	156.	c	157.	b	158.	a	159.	c	160.	b
161.	b	162.	b	163.	c	164.	c	165.	c	166.	b	167.	b	168.	c	169.	a	170.	c
171.	b	172.	d	173.	b	174.	b	175.	b	176.	d	177.	d	178.	b	179.	c	180.	d
181.	a	182.	b	183.	b	184.	c	185.	c	186.	d	187.	c	188.	b	189.	a	190.	c
191.	a	192.	c	193.	c	194.	a	195.	c	196.	b	197.	c	198.	a	199.	b	200.	c
201.	c	202.	a	203.	b	204.	a	205.	c	206.	d	207.	b	208.	d	209.	a	210.	c
211.	d	212.	b	213.	b	214.	c	215.	a	216.	b	217.	c	218.	c	219.	d	220.	a
221.	c	222.	d	223.	d	224.	c	225.	d	226.	d	227.	d	228.	b	229.	c	230.	c
231.	c	232.	d	233.	a	234.	a	235.	d	236.	b	237.	a	238.	a	239.	c	240.	d
241.	c	242.	a	243.	c	244.	b	245.	a	246.	a	247.	b	248.	c	249.	c	250.	d
251.	b	252.	d	253.	d	254.	b	255.	b	256.	d	257.	c	258.	d	259.	c	260.	b
261.	a	262.	a	263.	b	264.	c	265.	a	266.	a	267.	b	268.	b				

Solutions

Sol. 1. (c)

Class interval	Frequency(f)	(cf)
0-10	5	5
10-20	10	15
20-30	20	35
30-40	5	40
40-50	10	50
	N=50	

$$\therefore \frac{N}{2} = \frac{50}{2} = 25$$

Here, $f_k = 20$, $l = 20$

$h = 10$, $c = 15$

Now, median =

$$= 20 + \frac{10}{20}(25 - 15)$$

$$= 20 + \frac{10}{20} \cdot 10 = 20 + 5 = 25$$

Sol. 2. (b)

Class	(f)	(x)	(fx)
0-10	5	5	25
10-20	10	15	150
20-30	20	25	500
30-40	5	35	175
40-50	10	45	450
	N=50		$\Sigma fx = 1300$

$$\text{Now, Mean} = \frac{\Sigma(f \times x)}{N} = \frac{1300}{50} = 26$$

Solutions (for next eight):

Year	Male			Female			Total
	Urban	Rural	Total	Urban	Rural	Total	
1995	280	350	630	410	310	720	1350
1996	370	300	670	180	270	450	1120
1997	310	130	440	180	190	370	810
1998	400	280	680	290	80	370	1050
Total	1360	1060	2420	1060	850	1910	4330

Sol. 12. (a)

The total population for the year 1997 = 810

Sol. 13. (b)

The female urban population in the year 1995 = 410

Sol. 14. (c)

The total number of urban population in the year 1997 = 310 + 180 = 490

Sol. 15. (d)

The total population in the year 1998 = 1050

Sol. 16. (a)

The difference between the number of females and the number of males in the year 1995 = $(720 - 630) = 90$

Sol. 17. (c)

In 1997 the male population is minimum which is 440.

Sol. 18. (a)

In 1995 the female population is maximum, which is 720.

Sol. 19. (a)

The percentage of rural male population (over the whole population) in the year 1998 = $\frac{280}{1060} \times 100$

Sol. 3. (b)

Class Interval	f	cf
0.5-5.5	3	3
5.5-10.5	7	10
10.5-15.5	6	16
15.5-20.5	5	21
	21	50

$$\text{Here, } N = 21 \Rightarrow \frac{N}{2} = \frac{21}{2} = 10.5$$

$\therefore$ Median lies in 10.5-15.5

$$\therefore \text{Median} = 10.5 + \frac{10.5 - 10}{6} \times 5$$

$$= 10.5 + 0.417 = 10.917$$

Here, modal class is 5.5-10.5

Hence, median is not contained in the modal class

Firstly frequency is increasing then decreasing so graph will be skew symmetric bell shape.

Sol. 4. (c)

We know that, when two lines are coincident linearly, then $r = \pm 1$ and also, if $r \in [-1, 1]$, then there is a relation measure of the linear relationship between the variables.

Sol. 5. (b)

If the values of a set are measured in cm, then the variance has unit cm^2 .

Sol. 6. (c)

The cumulative frequency curve of statistical data commonly called Ogive.

Sol. 7. (d)

$$110 = \frac{x + 9 \times 76}{10} \Rightarrow x = 416$$

Sol. 8. (c)

G.M. of 2, 4, 8, 16, 32

$$= (2 \cdot 4 \cdot 8 \cdot 16 \cdot 32)^{\frac{1}{5}} = (2^{15})^{\frac{1}{5}} = 8$$

Sol. 9. (a)

Given values of variates 2, 3, 4, 2, 5, 4, 2. The higher frequency value, which takes variate is mode = 2.

Sol. 10. (c)

$$\text{Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{(12 + 40 + 98 + 216 + 192 + 112 + 90)}{(2 + 4 + 7 + 12 + 8 + 4 + 3)}$$

$$= \frac{760}{40} = 19$$

Sol. 11. (b)

We know that, $\text{Var}(\lambda x) = \lambda^2 \text{Var}(x)$

... (i)

Given data is. $x = 2, 4, 5, 6, 17$ and its variance $\text{Var}(x) = v$.

Now, multiply by 2 in above data numbers;

$x = 4, 8, 10, 12, 34$

then, its variance, $\text{Var}(2x) = (2)^2 \text{Var}(x) = 4v$

$$= \frac{2800}{106} = \frac{1400}{53} = 26.4\% = \frac{80}{3}\% (\text{approx})$$

Sol. 20. (c)

Employment is allocated maximum funds.

Sol. 21. (a)

$$\text{Money allocated to education} = \frac{30^\circ}{360^\circ} \times 36000 = 3000$$

Sol. 22. (b)

Money allocated to both agriculture and employment =

$$\left(\frac{90^\circ + 120^\circ}{360^\circ} \right) \times 36000 = \frac{210}{360} \times 36000 = 21000$$

Sol. 23. (c)

$$\text{Required value of money} = \left(\frac{75^\circ - 30^\circ}{360^\circ} \right) \times 36000$$

$$= \frac{45}{360} \times 36000 = 4500$$

Sol. 24. (b)

The geometric mean of the given observations

$$= (10 \cdot 40 \cdot 60)^{\frac{1}{3}} = (24 \times 1000)^{\frac{1}{3}}$$

$$= (3)^{\frac{1}{3}} (8 \times 1000)^{\frac{1}{3}}$$

$$= (3)^{\frac{1}{3}} (2 \times 10) = 20 \sqrt[3]{3}$$

Sol. 25. (a)

First, we arranged given observation in ascending order 2, 3, 4, 5, 6, 7, 9

Total terms, $n = 7$ (odd)

$$\therefore \text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ term}$$

$$= \left(\frac{7+1}{2} \right)^{\text{th}} \text{ term} = 4^{\text{th}} \text{ term} = 5$$

Sol. 26. (c)

Given mean of 7 observation = 0

$$\Rightarrow \frac{\sum_{i=1}^7 x_i}{7} = 10 \Rightarrow \sum_{i=1}^7 x_i = 70 \quad \dots (i)$$

and mean of 3 observations = 5

$$\Rightarrow \frac{\sum_{i=1}^3 x_i}{3} = 5 \Rightarrow \sum_{i=1}^3 x_i = 15 \quad \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$\sum_{i=1}^7 x_i + \sum_{i=1}^3 x_i = 70 + 15 \Rightarrow 70 + 15 = \sum_{i=1}^{10} x_i = 85$$

$$\therefore \text{Mean of 10 observations} = \frac{\sum_{i=1}^{10} x_i}{10} = \frac{85}{10} = 8.5$$

Sol. 27. (c)

Let A, B, C be the section of class having 30, 30 and 40 students respectively.

Also given the student of each section securing the arithmetic means of the marks 72.2, 69.0 and 64.1 respectively.

Now, the total marks secure by the students of section A = $30 \times 72.2 = 2166$

The total marks secure by the students of section B = $30 \times 69 = 2070$

and the total marks secure by the students of section C = $40 \times 64.1 = 2564$

So, the arithmetic mean of marks of all the students of three sections

$$= \frac{(2166 + 2070 + 2564)}{100}$$

$$= \frac{6800}{100} = 68$$

Sol. 28. (d)

Let a and b two observations. Then,

I. Arithmetic mean (AM) = $\left(\frac{a+b}{2}\right)$

Now, we multiply by c in every observation, then

$$AM = \frac{ac+bc}{2} = c\left(\frac{a+b}{2}\right)$$

II. Geometric mean (GM) = $\sqrt{ab}$

Now, we take ac and bc,

Then GM = $\sqrt{ac \cdot bc} = c\sqrt{ab}$

III. Harmonic mean, (HM) = $\frac{2ab}{a+b}$

Now, we take ac, bc

$$HM = \frac{2(ac) \cdot (bc)}{(ac+bc)} = \frac{2c^2ab}{c(b+a)} = c\left(\frac{2ab}{a+b}\right)$$

IV. Median let a, b, d, f and h be five observations, then total terms = $4 + 1 = 5 = n$

$$\text{Median} = \left(\frac{n+1}{2}\right)\text{th term} = \left(\frac{5+1}{2}\right)\text{th term}$$

$$= 3^{\text{rd}} \text{ term} = d$$

Now we take ac, bc, dc, fc, hc

Then, Median = $\left(\frac{5+1}{2}\right)\text{th term} = 3^{\text{rd}} \text{ term} = c \cdot d$

and let a, b, d and f be four observations, then total terms = $4 = n$

$$\text{Median} = \left(\frac{n}{2}\right)\text{th term} + \left(\frac{n}{2} + 1\right)\text{th term}$$

$$= 2^{\text{nd}} \text{ term} + 3^{\text{rd}} \text{ term} = (b+d)$$

Now, we take, ac, bc, dc, fc

$$\text{Then, Median} = \left(\frac{n}{2}\right)\text{th term} + \left(\frac{n}{2} + 1\right)\text{th term}$$

$$= 2^{\text{nd}} \text{ term} + 3^{\text{rd}} \text{ term}$$

$$= bc + dc = c(b+d)$$

Hence, option (d) is correct.

Sol. 29. (c)

A.M. of two numbers is 10, i.e. sum of numbers is 20.

G.M. of two numbers is 8, i.e. product of numbers is 64.

two numbers whose sum is 20 and product is 64 are 16, 4.

difference of numbers is 12.

Sol. 30. (c)

The given observation are,

20, 20, 20 21, 21, 21, 21, 21, 22, 22, 22, 22, 22, 22, 22, 23, 23, 23, 23, 23, 24, 24, 25

Mode = higher frequency of observation = 22 (7 times)

Sol. 31. (a)

Here, statement III is wrong because construction of a frequency distribution is based on data which are both discrete as well as continuous.

Sol. 32. (b)

Sol. 33. (b)

In increasing order 2, 2, 3, 3, 4, 5, 7, 9, 10

Middle value = median = 4

Sol. 34. (c)

$$\text{mean} = \frac{(1(1) + 2(2) + 3(3) + \dots + 16(16))}{1 + 2 + 3 + 4 + \dots + 16}$$

$$= \frac{16(16+1)(32+1)}{16(16+1)} = 11$$

$$\text{mean} = \frac{6}{\frac{16(16+1)}{2}} = 11$$

Sol. 35. (c)

we know that for two numbers $G^2 = AH$

$$(10)^2 = A(8)$$

$$A = 12.5$$

Sol. 36. (d)

According to question,

$$\sum_{i=1}^{20} (x_i - 30) = 2 \quad (\text{given})$$

$$\Rightarrow \sum_{i=1}^{20} x_i - 600 = 2 \Rightarrow \sum_{i=1}^{20} x_i = 602$$

$$\therefore \text{Mean} = \frac{\sum_{i=1}^{20} x_i}{20} = \frac{602}{20} = 30.1$$

Sol. 37. (d)

$$\therefore \frac{7+9+11+13+15}{5} = \frac{55}{5} = 11$$

$$\text{Now, SD} = \sqrt{\frac{(7-11)^2 + (9-11)^2 + (11-11)^2 + (13-11)^2 + (15-11)^2}{5}}$$

$$= \sqrt{\frac{16+4+0+4+16}{5}} = \sqrt{8} = 2.8 (\text{approx})$$

Sol. 38. (d)

Let observations are $x_1, x_2, \dots, x_{10}$

$$\text{Given, } \frac{x_1 + x_2 + x_3 + \dots + x_{10}}{10} = 5$$

$$\Rightarrow x_1 + x_2 + x_3 + \dots + x_{10} = 50$$

Again, according to the question,

New mean

$$= \frac{[(x_1 + 2) + (x_2 + 2) + (x_3 + 2) + \dots + (x_{10} + 2)] \times 3}{10}$$

$$= \frac{[(x_1 + x_2 + x_3 + \dots + x_{10}) + 20] \times 3}{10}$$

$$= \frac{(50 + 20) \times 3}{10} = \frac{70 \times 3}{10} = 21$$

Sol. 39. (a)

Median of 27 observations = 18

$$\text{Then, } \left(\frac{27+1}{2}\right)\text{th observation} = 18$$

and 14th observation = 18

Then, observation whose value is 16 comes before 18 i.e., 14th observation.

Now, number of observation is even 30 observation.

$$\text{Median} = \frac{\left(\frac{30}{2}\right)\text{th} + \left(\frac{30}{2} + 1\right)\text{th}}{2}$$

$$= \frac{15\text{th observation} + 16\text{th observation}}{2}$$

$$= \frac{18+18}{2} = 18$$

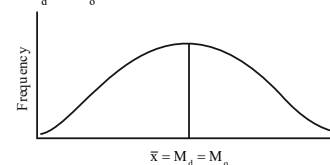
Sol. 40. (d)

Frequency curve may be symmetrical,

Positive skew and negative skew

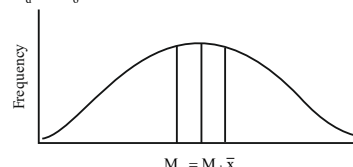
For symmetry $\Rightarrow \text{Mean} = \text{Median} = \text{Mode}$

$$\Rightarrow \bar{x} = M_d = M_o$$



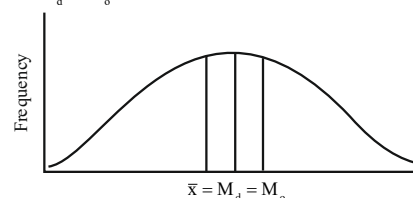
For positive skew. Mean > Median > Mode

$$\bar{x} > M_d > M_o$$



For negative skew. Mean < Median < Mode

$$\bar{x} < M_d < M_o$$



Sol. 41. (a)

sum of n odd natural numbers = n^2

Sol. 42. (c)

algebraic sum about mean is always zero.

Sol. 43. (c)

transport = 10%

expenditure = 900

Sol. 44. (c)

$$\text{coefficient of variation} = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{8}{40} \times 100 = 20\%$$

Sol. 45. (d)

others are measure of central tendency.

Sol. 46. (d)

intersection point of these lines is $(-3, -2)$

Sol. 47. (a)

G.M. of 1, 2, 4, 8, 16, ..., 2^n

$$= (1 \cdot 2 \cdot 4 \cdot 8 \cdot 16 \cdot \dots \cdot 2^n)^{\frac{1}{n+1}} = (2^{0+1+2+3+4+\dots+n})^{\frac{1}{n+1}}$$

$$= \left(2^{\frac{n(n+1)}{2}}\right)^{\frac{1}{n+1}} = 2^{\frac{n}{2}}$$

Sol. 48. (b)

First, we arrange the given data in ascending order, we get 3, 6, 6, 7, 7, 7, 8, 9, 9, 10, 10, 10, 12

Total terms, $n = 13$ (odd)

$$\therefore \text{Median} = \left(\frac{n+1}{2}\right)\text{th term}$$

$$= \left(\frac{13+1}{2}\right)\text{th term}$$

$$= 7^{\text{th}} \text{ term} = 8$$

Sol. 49. (d)

Both variance and standard deviation are not the measures of variability in the population.

Also, relation between variance and standard deviations, $\sqrt{\text{Variance}} = \text{Standard deviation}$

$$\Rightarrow \sqrt{\sigma} = \text{SD}$$

$$\Rightarrow \sigma = (\text{SD})^2$$

Hence, both statements are incorrect.

Sol. 50. (a)

Variance is always independent of the change of origin but not scale.

eg., $\text{Var}(ax + b) = a^2 \text{V}(x)$

Sol. 51. (a)

Given observations are 5, 5, 5, 5 and 5.

$$\text{Mean } (\bar{x}) = \frac{5+5+5+5+5}{5} = \frac{25}{5} = 5$$

$$\text{Now, SD} = \sqrt{\frac{\sum_{i=1}^5 (x_i - \bar{x})^2}{N}}$$

$$= \sqrt{\frac{(5-5)^2 + (5-5)^2 + (5-5)^2 + (5-5)^2 + (5-5)^2}{5}}$$

$$= \sqrt{\frac{0+0+0+0+0}{5}} = 0$$

Sol. 52. (c)

Class Interval	Frequency
0-10	14
10-20	x
20-30	27
30-40	y
40-50	15

Given that,

Sum of frequencies = 100

$$\Rightarrow 14 + x + 27 + y + 15 = 100$$

$$\Rightarrow x + y + 57 = 100$$

$$\Rightarrow x + y = 43 \quad \dots(i)$$

For mode, $f_m = 27$, $f_1 = x$ and $f_2 = y$, $I_1 = 20$, $h = 10$

Clearly, 20-30 is the modal class

Since, mode lies between 20-30

$$\therefore \text{Mode} = I_1 + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$\text{Given, } 25 = 20 + \frac{27 - x}{54 - x - y} \times 10$$

$$\Rightarrow \frac{5(270 - 10x)}{54 - x - y}$$

$$\Rightarrow 270 - 10x = -5x - 5y + 270$$

$$\Rightarrow 5x - 5y = 0 \Rightarrow x = y \quad \dots(ii)$$

From Eqs. (i) and (ii)

$$2x = 43 \Rightarrow x = \frac{43}{2} = 21.5$$

$$\therefore x = y = 21.5$$

Sol. 53. (a)

Given that, average marks obtained by the student in a class $(\bar{x}_{BG}) = 43$

Total number of boys $(n_B) = 25$

Average marks obtained by boys $(\bar{x}_B) = 40$

Average marks obtained by girls $(\bar{x}_G) = 48$

Let the total number of girls = n_G

Now, by formula,

$$\bar{x}_{BG} = \frac{n_B \cdot \bar{x}_B + n_G \cdot \bar{x}_G}{n_B + n_G}$$

$$\Rightarrow 43 = \frac{25 \cdot 40 + n_G \cdot 48}{25 + n_G}$$

$$1075 + 43 \cdot n_G = 1000 + 48 \cdot n_G$$

$$\Rightarrow 75 = 5n_G$$

$$\Rightarrow n_G = 15$$

$\therefore$ Required number of girls = 15

Sol. 54. (c)

We know that,

$$\tan \theta = \frac{1-r^2}{|r|} \cdot \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

Where θ is the angle between the two regression lines? Given that, two lines of regression are perpendicular.

i.e., $(\theta = 90^\circ)$

$$\therefore \tan 90^\circ = \frac{1-r^2}{|r|} \cdot \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2} = \infty = \frac{1}{0}$$

$$\Rightarrow |r| \cdot \sigma_x^2 + \sigma_y^2 = 0$$

$$\Rightarrow |r| = 0 \quad (\because \sigma_x^2 + \sigma_y^2 \neq 0)$$

$$\therefore r = 0$$

i.e., Correction coefficient $(r) = 0$

Sol. 55. (d)

$$\text{mean} = (30+55+75+90+50+60+39)/7$$

$$= 57 \text{ less numbers 4}$$

Sol. 56. (a)

$$2A + G^2 = 27$$

$$2A + AH = 27$$

$$2A + 4A = 27$$

$$A = 4.5$$

so G will be 18

If $A = 4.5$ then sum of numbers is 9

If $G^2 = 18$ then product of numbers is 18

two numbers whose sum 9 and product 18 are 6 and 3

Sol. 57. (b)

Give that,

Mean of 20 observation = 15

$$\therefore \text{Sum of 20 observations} = 20 \times 15 = 300$$

$$\therefore \text{Sum of actual (correct) observations}$$

$$= 300 - (3+6) + (8+4) = 300 - 9 + 12 = 303$$

$$\therefore \text{correct mean} = \frac{303}{20} = 15.15$$

Sol. 58. (c)

Arithmetic mean of the squares of the first 'n' natural numbers

$$= \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n}$$

$$= \frac{n(n+1)(2n+1)}{6 \times n} = \frac{(n+1)(2n+1)}{6}$$

Sol. 59. (c)

$$\text{Let } b_{yx} > 1 \Rightarrow \frac{1}{b_{xy}} < 1 \quad \dots(i)$$

We have, $b_{yx} \cdot b_{xy} = r^2 \leq 1$

$$\Rightarrow b_{yx} \leq \frac{1}{b_{xy}} < 1 \quad \dots(\text{From Eq. (i)})$$

Hence, if one of the regression coefficient is greater than one, the other must be less than one.

So, both statements are true.

Sol. 60. (c)

Sol. 61. (d)

Sol. 62. (a)

$$\text{Variance } (V) = \frac{n^2 - 1}{12} = \frac{120}{12} = 10$$

Sol. 63. (c)

Sol. 64. (c)

Sol. 65. (c)

variance will be multiply by 4

Sol. 66. (a)

I. We know that, variance is not dependent on change of origin i.e., independent on change of origin.

So, if every x_i is increased by 2, the variance of the new set of numbers is not change i.e., V.

Sol. 67. (b)

Mean of first 20 natural numbers

$$= \frac{\text{sum of the squares first of 20 natural numbers}}{\text{Number of observations}}$$

$$= \frac{(1^2 + 2^2 + 3^2 + \dots + 20^2)}{20}$$

$$\left[\because \frac{n(n+1)(2n+1)}{6} \right]$$

$$= \frac{1}{20} \times \frac{20(20+1)(20 \times 2 + 1)}{6}$$

$$= \frac{21 \times (40+1)}{6}$$

$$= \frac{21 \times 41}{6} = \frac{7 \times 41}{2} = \frac{287}{2} = 143.5$$

$\therefore$ required mean = 143.5

Sol. 68. (c)

Given that, p, q, r, s and t are five numbers,

$\therefore$ Average of p, q, and r = 5

$$\Rightarrow \frac{p+q+r}{3} = 5$$

$$\Rightarrow p+q+r = 15 \quad \dots(i)$$

and average of s and t = 10

$$\Rightarrow \frac{s+t}{2} = 10$$

$$\Rightarrow s+t = 20 \quad \dots(ii)$$

Now, average of all five numbers

$$= \frac{p+q+r+s+t}{5} = \frac{(p+q+r) + (s+t)}{5}$$

$$= \frac{15+20}{5} \quad [\text{From Eq.(i) and (ii)}]$$

$$\frac{35}{5} = 7$$

Solutions (for next three): Given frequency distribution is

(x_i)	0	1	2	3	4	5	6	7
(f_i)	14	21	25	43	51	40	39	12

Sol. 69. (a)

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$$

$$= \frac{0 \times 14 + 1 \times 21 + 2 \times 25 + 3 \times 43 + 4 \times 51 + 5 \times 40 + 6 \times 39 + 7 \times 12}{(14+21+25+43+51+40+39+12)}$$

$$= \frac{(0+21+50+129+204+200+234+84)}{245}$$

$$= \frac{922}{245} = 3.76$$

$$\frac{922}{245}$$

Sol. 70. (b)

Number of calls (x_i)	Frequency (f_i)	Cumulative Frequency (F)
0	14	14
1	21	35
2	25	60
3	43	103
4	51	154
5	40	194
6	39	233
	$N = \sum f_i =$	245

$$\text{Here, } \frac{N}{2} = \frac{245}{2} = 122.5$$

The cumulative frequency 154 which is equal or just greater than $N/2$

∴ Required median = Value of the variable corresponding to the cumulative frequency 154 = 4

Sol. 71. (b)

We see that in the frequency distribution the higher frequency is 51.

∴ Required mode = Value of variable corresponding to the higher frequency = 4.

Sol. 72. (c)

Given that, n = Total number to tosses = 18 and X = Number of scores which exceed 4 in 18 tosses of a symmetrical die = {5,6}

$$\Rightarrow n(X) = 2$$

$$\therefore p = \frac{n(X)}{n(s)} = \frac{2}{6} = \frac{1}{3}$$

$$\text{and } q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3} \quad (\because p+q=1)$$

Now, arithmetic mean of X = np

$$= 18 \times \frac{1}{3} = 6$$

and standard deviation of

$$X = \sqrt{\text{Variance } X}$$

$$= \sqrt{npq} = \sqrt{18 \times \frac{1}{3} \times \frac{2}{3}}$$

$$= \sqrt{4} = 2$$

Hence, both statements are correct.

Sol. 73. (c)

Sol. 74. (b)

Here, $n_1 = 100$, $\bar{X}_{100} = 50$

and $n_2 = 150$ and $\bar{X}_{150} = 40$

∴ combined mean of all 250 items i.e.,

$$\bar{X}_{250} = \frac{n_1 \cdot \bar{X}_{100} + n_2 \cdot \bar{X}_{150}}{n_1 + n_2}$$

$$= \frac{100 \times 50 + 150 \times 40}{100 + 150} = \frac{5000 + 6000}{250} = \frac{11000}{250} = 44$$

Sol. 75. (c)

We know that, If n_1 and n_2 are the sizes $\bar{X}_{100}$, $\bar{X}_{150}$ are the means of σ_{100} , σ_{150} are the standard deviation of the series, then the standard deviation of the combined series is:

$$\sigma = \sqrt{\frac{n_1(\sigma_{100}^2 + d_1^2) + n_2(\sigma_{150}^2 + d_2^2)}{n_1 + n_2}} \quad \dots(i)$$

Where, d_1

$$= \bar{X}_{100} - \bar{X}_{250} \text{ and } d_2 = \bar{X}_{150} - \bar{X}_{250}$$

$$\text{Here, } d_1 = 50 - 44 = 6$$

$$\Rightarrow d_1^2 = 36$$

$$\text{and } d_2 = 40 - 44 = -4$$

$$\Rightarrow d_2^2 = 16$$

From Eq. (i)

Combined standard deviation of all 250 items i.e.,

$$\begin{aligned} \sigma_{250} &= \sqrt{\frac{100\{(5)^2 + 36\} + 150\{(6)^2 + 16\}}{100 + 150}} \\ &= \sqrt{\frac{100(25 + 36) + 150(36 + 16)}{250}} \\ &= \sqrt{\frac{10 \times 61 + 15 \times 52}{5}} = \sqrt{\frac{610 + 780}{5}} = \sqrt{\frac{1390}{5}} \\ &= \frac{37.28}{5} = 7.456 \approx 7.5 \end{aligned}$$

Sol. 76. (c)

Variance of all the 250 items = $(\sigma_{250})^2$

$$= (7.456)^2 = 55.59 \approx 55.6$$

Sol. 77. (c)

Coefficient of correlation

$$= \sqrt{\left(-\frac{3}{2}\right) \times \left(-\frac{1}{6}\right)} = -0.5$$

Sol. 78. (b)

Mean deviation about the mean

$$= \frac{\sum |x_i - \bar{x}|}{N}$$

$$\text{Here, } \bar{x} = \frac{4 + 7 + 8 + 9 + 10 + 12 + 13 + 17}{8} = 10$$

∴ Mean deviation about mean

$$\begin{aligned} & \frac{|4-10| + |7-10| + |8-10| + |9-10| + |10-10|}{8} \\ & + \frac{|12-10| + |13-10| + |17-10|}{8} \\ & = \frac{6 + 3 + 2 + 1 + 0 + 2 + 3 + 7}{8} = \frac{24}{8} = 3 \end{aligned}$$

Sol. 79. (c)

Let $x_1, x_2, \dots, x_{20}$ be the given observations.

We have,

$$\frac{1}{20} \sum_{i=1}^{20} (x_i - \bar{x})^2 = 5$$

To find variance of $2x_1, 2x_2, 2x_3, \dots, 2x_{20}$. Let $\bar{X}$ denotes the mean of new observation.

$$\text{Clearly, } \bar{x} = \frac{\sum_{i=1}^{20} 2x_i}{20} = \frac{2 \sum_{i=1}^{20} x_i}{20} = 2\bar{x}$$

Now, variance of new observation

$$= \frac{1}{20} \sum_{i=1}^{20} (2x_i - \bar{x})^2$$

$$= \frac{1}{20} \sum_{i=1}^{20} (2x_i - 2\bar{x})^2$$

$$= \frac{1}{20} \sum_{i=1}^{20} 4(x_i - \bar{x})^2$$

$$= 4 \left(\frac{1}{20} \sum_{i=1}^{20} (x_i - \bar{x})^2 \right)$$

$$= 4 \times 5 = 20$$

Sol. 80. (a)

I. It is true that, the histogram is a suitable representation of a frequency distribution of a continuous variable. Hence, statement I is correct.

II. We know that, the area of histogram is proportional to the frequency, so it is not correct statement.

Sol. 81. (d)

It is obvious that, $\bar{x} < \bar{z} < \bar{y}$

Sol. 82. (d)

Angle between the regression lines will be

$$\tan \theta = \left\{ \left(\frac{1-r^2}{r} \right) \left(\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right) \right\}$$

$$\Rightarrow \tan \frac{\pi}{2} = \left(\frac{1-r^2}{r} \right) \left(\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right)$$

$$\Rightarrow r \cdot (\sigma_x^2 + \sigma_y^2) = 0$$

$$\therefore r = 0$$

Sol. 83. (a)

Harmonic mean is used in construction of index numbers.

Sol. 84. (c)

Both Statements are true

Sol. 85. (b)

Only Statement II is correct.

Sol. 86. (d)

Let the numbers are x_1, x_2, x_3, x_4 and x_5

Then,

$$\frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = 30$$

$$\therefore x_1 + x_2 + x_3 + x_4 + x_5 = 150 \dots(i)$$

Now, suppose x_1 is excluded,

$$\frac{x_2 + x_3 + x_4 + x_5}{4} = 28 \quad \dots(\text{given})$$

$$\Rightarrow x_2 + x_3 + x_4 + x_5 = 112 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$x_1 = 150 - 112 = 38$$

Sol. 87. (a)

The median of grouped data is the x-coordinate of the point of intersection of 'less than' and 'more than' ogive curve.

Sol. 88.

coefficient of correlation is independent from change of origin and change of scale, but depends upon change of sign. sign of X observation is changing from X to -X so now coefficient of correlation will be -0.6.

Sol. 89. (c)

mean multiplied by 4

and variance multiplied by $(2)^2$

so new mean and variance will be 8, 8

Sol. 90. (c)

$$\text{sum of cube of } n \text{ natural numbers} = \left[\frac{n(n+1)}{2} \right]^2$$

$$\text{mean} = \left[\frac{n(n+1)^2}{4} \right]$$

Sol. 91. (c)

$$G_1 = [x_1 \times x_2 \times x_3 \dots x_n]^{1/n}$$

$$G_2 = [y_1 \times y_2 \times y_3 \dots y_n]^{1/n}$$

$$\frac{G_1}{G_2} = \left[\frac{x_1}{y_1} \times \frac{x_2}{y_2} \times \frac{x_3}{y_3} \dots \frac{x_n}{y_n} \right]^{1/n}$$

$$\frac{G_1}{G_2} \text{ is the G.M. of } \frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, \dots, \frac{x_n}{y_n}$$

Sol. 92. (b)

Coefficient of correlation

$$= \sqrt{(-0.64) \times (-0.36)} = -0.48$$

Sol. 93. (c)

$$n = 20, \sum x_i = 1000 \text{ and } \sum x_i^2 = 84000$$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{x} = \frac{1000}{20}$$

$$\bar{x} = 50$$

$$\text{Variance } (\sigma^2) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$= \frac{84000}{20} - (50)^2$$

$$= 4200 - 2500$$

$$\sigma^2 = 1700$$

Sol. 94. (a)

10, 9, 21, 16, 24

$$\bar{x} = \frac{10 + 9 + 21 + 16 + 24}{5}$$

$$\bar{x} = \frac{80}{5}$$

$$\bar{x} = 16$$

x	x - $\bar{x}$
10	6
9	7
21	5
16	0
24	8
	26

$$\text{Mean Deviation} = \frac{\sum |x - \bar{x}|}{n} = \frac{26}{5}$$

$$\text{Mean Deviation} = 5.2$$

Sol. 95. (d)

From the given values,

$$\bar{X} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

$$x_1 + x_2 + x_3 + \dots + x_n = n\bar{X}$$

Now, new sum of the data when we replace x_2 by λ i.e.,

$$\text{New sum} = n\bar{X} - x_2 + \lambda$$

$$\text{New Mean} = \frac{\text{New sum}}{\text{Total no. data}}$$

$$\text{New Mean} = \frac{n\bar{X} - x_2 + \lambda}{n}$$

Sol. 96. (d)

Given data 3, 5, 1, 6, 5, 9, 5, 2, 8, 6

$$\text{Mean} = \bar{x} = \frac{\sum x_i}{n}$$

$$\therefore \bar{x} = \frac{3+5+1+6+5+9+5+2+8+6}{10} = 5$$

$$x = 50/10$$

$$x = 5$$

$$1, 2, 3, 5, 5, 5, 6, 6, 8, 9$$

$$n = 10 \text{ (even)}$$

Median

$$= \frac{1}{2} \left[\left(\frac{n}{2} \right) \text{th term} + \left(\frac{n}{2} + 1 \right) \text{th term} \right]$$

or $y =$

$$\frac{1}{3} \left[\left(\frac{10}{2} \right) \text{th term} + \left(\frac{10}{2} + 1 \right) \text{th term} \right] \therefore \left\{ \begin{array}{l} 5 \\ 6 \end{array} \right\} =$$

$$\frac{1}{2} [5 \text{th term} + 6 \text{th term}] = \frac{1}{2} [5 + 5]$$

$$\therefore y = 5$$

$$\text{Mode} = z = 5$$

Sol. 97. (c)

Both statements are correct

Sol. 98. (a)

let any value of x, y, z

$$\text{let } x = y = z = 2$$

by substituting these values our data becomes 1, 1, 1

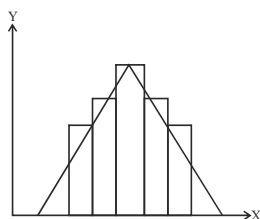
GM of this data is 1

Sol. 99. (d)

I. In a symmetric distribution, values of variables occur at regular frequencies, hence mean and median are equal in symmetric distribution.

II. Range = Largest Value - Smallest value

III.



Clearly, sum of area of rectangles is equal to area under frequency polygon.

Sol. 100. (d)

$$\text{Coefficient of variation} = \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

$$CV(\text{Group A}) = \frac{10}{22} \times 100 = 45.45$$

$$CV(\text{Group B}) = \frac{12}{23} \times 100 = 52.17$$

So, Group A is less variable than group B because Group A's coefficient of variation is smaller.

Sol. 101. (b)

For grouped frequency distribution class-intervals should be exhaustive. Class-intervals are generally equal in width but this might not be the case always.

Sol. 102. (b)

Age	Frequency f_i	Mid-values x_i	$f_i x_i$
15-25	2	20	40
25-35	4	30	120
34-35	6	40	240
45-55	5	50	250
55-65	3	60	180
	$\Sigma f = 20$		$\Sigma f_i x_i = 830$

$$\therefore \text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = \frac{830}{20} = 41.5$$

Sol. 103. (b)

Given, $\text{Cov}(x, y) = 30$

$$\sigma_x^2 = 25 \Rightarrow \sigma_x = 5$$

$$\text{and } \sigma_y^2 = 144 \Rightarrow \sigma_y = 12$$

$\therefore$ Correlation coefficient

$$r(x, y) = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} = \frac{30}{5 \times 12} = \frac{30}{60} = 0.5$$

Sol. 104. (d)

Given,

Scores 10, 5, 8, 16, 18, 20, 8, 10, 16, 20, 18, 11, 16, 14, 12

$$\text{Mean } (\bar{x}) = \frac{\sum x_i}{n} = \frac{202}{15} = 13.46$$

Scores in ascending order,

5, 8, 8, 10, 11, 12, 14, 16, 16, 16, 18, 18, 20, 20

Here $n = 15$ (odd)

Median

$$= \left(\frac{n+1}{2} \right) \text{th} = \left(\frac{15+1}{2} \right) \text{th} = \left(\frac{16}{2} \right) \text{th}$$

$$= 8^{\text{th}} = 14$$

If score 16 change as 18 the mean

$$(\bar{x}) = \frac{208}{15} = 13.87$$

Median = 14

Mode = 18

So, mean will be changed and median will not be changed and mode will be changed.

Sol. 105. (b)

Given, $\Sigma x = 130$, $\Sigma y = 220$, $\Sigma x^2 = 2288$, $\Sigma y^2 = 5506$ and $\Sigma xy = 3467$

We know that, regression line of y on x is $y = a + bx$

To determine the values of a and b , we will solve the normal equations.

$$\Sigma y = na + b \Sigma x \text{ and } \Sigma xy = a \Sigma x + b \Sigma x^2$$

Substituting the required value, the normal equations.

$$220 = 10a + b(130)$$

$$\Rightarrow a + 13b = 22 \quad \dots(i)$$

$$\text{and } 3467 = a(130) + b(2288)$$

$$\Rightarrow 130a + 2288b = 3467 \quad \dots(ii)$$

Multiply equation (i) by 130 and subtracting from equation (ii), we get

$$130a + 2288b = 3467$$

$$130a + 1690b = 2860$$

$$598b = 607$$

$$\Rightarrow b = \frac{607}{598} = 1.02$$

From equation (i), we get

$$a + 13 \times 1.02 = 22$$

$$\Rightarrow a + 13.26 = 22$$

$$\Rightarrow a = 8.74$$

So, equation of regression of line

$$y = 8.74 + 1.02x$$

Sol. 106. (c)

$$\therefore \text{var}(x-y) = \text{var}(x) + \text{var}(-y) = \text{var}(x) + \text{var}(y)$$

$$= \sigma_x^2 + \sigma_y^2$$

$$\text{and var}(x+y) = \text{var}(x) + \text{var}(y) = \sigma_x^2 + \sigma_y^2$$

$$\text{cov}(x+y, x-y) = \text{cov}(x, x) - \text{cov}(x, y) + \text{cov}(y, x) - \text{cov}(y, y)$$

$$= \text{var}(x) - \text{var}(y) = \sigma_x^2 - \sigma_y^2$$

$\therefore$ Correlation coefficient

$$= \frac{\text{cov}(x+y, x-y)}{\sqrt{\text{var}(x+y)} \sqrt{\text{var}(x-y)}}$$

$$\frac{\sigma_x^2 - \sigma_y^2}{\sigma_x^2 + \sigma_y^2}$$

Sol. 107. (b)

Sol. 108. (b)

The line of regression of y on x : $4y - 15 + 410 = 0$

$$\Rightarrow y = \frac{15}{4}x - \frac{410}{4}$$

$$\therefore b_{yx} = \frac{15}{4}$$

The line of regression of x on y : $30x - 2y - 325 = 0$

$$\Rightarrow x = \frac{2y}{30} + \frac{325}{30}$$

$$\therefore b_{xy} = \frac{1}{15}$$

$$r^2 = b_{yx} \cdot B_{xy} = \frac{15}{4} \times \frac{1}{15} = \frac{1}{4}$$

$$\Rightarrow r = \pm \frac{1}{2}$$

Since both the coefficient of regression are positive, so $r = +\frac{1}{2}$

Sol. 109. (c)

The angle corresponding to the second class

$$= \frac{360^\circ}{100} \times 40 = 144^\circ$$

Sol. 110. (a)

Range is the simplest but crude measure of dispersion and is highly affected by the existence of extreme value.

Sol. 111. (c)

The empirical relationship between a Mean deviation and standard (Quartile) deviation is: $4 \times \text{S.D.} = 5 \times \text{M.D.}$ (Learn)

Sol. 112. (d)

Data can be represented in the Textual, Tabular or Graphical form.

Sol. 113. (a)

The intersecting point of less than ogive and more than ogive of a statistical data is the median of that data.

Sol. 114. (a)

$$r^2 = -\frac{1}{2} \times -\frac{1}{2} \times -\frac{1}{8} = \frac{1}{16} \Rightarrow r = \pm \frac{1}{4}$$

since both the coefficients of regressions are negative, so, $r = -\frac{1}{4}$

Sol. 115. (b)

Since 35 and 40 both lie after 33, therefore the median will remain same when 40 is replaced by 35.

But the sum of all observations is changed as $32 \times 5 - 40 + 35 = 31 \times 5$

$\therefore$ New mean = 31

Thus, the median remains the same but the mean will decrease.

Sol. 116. (d)

The initial variance = 5

$$\text{or } V = (\text{S.D.})^2 = 5 \quad \dots(i)$$

Now, when each observation is multiplied by 3, the S.D. also gets multiplied by 3, i.e.

New S.D. = 3 (S.D)

$\therefore$ New variance

$$= [\text{New (S.D.)}]^2 = [3(\text{S.D.})]^2 = 9 \times (\text{S.D.})^2$$

$$9 \times 5 = 45 \quad \{\text{From (i)}\}$$

Sol. 117. (b)

The mean of 100 observations = 20

$$\Rightarrow \text{Sum of 100 observations} = 20 \times 100 = 2000$$

New sum after omitting incorrect observations = $2000 - 21 - 21 - 18 - 20 = 1920$

$$\therefore \text{New mean} = \frac{1920}{96} = 20$$

Sol. 118. (a)

According to the question,

$$B + G = 150 \quad \dots(i)$$

$$70 \times B + 55 \times G = 150 \times 60$$

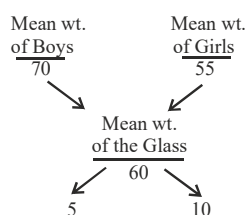
$$\text{or } 14B + 11G = 1800$$

Multiplying equation (i) by 11 and

Subtracting it from equation (ii),

$$3B = 150$$

$$\Rightarrow B = 50$$



Shortcut method:

The ratio of boys to girls = $5:10 = 1:2$

$$\Rightarrow x + 2x = 150$$

$$\Rightarrow x = 50$$

$\therefore$ Number of boys = 50

and number of girls = 100

Sol. 119. (b)

Statement 2 and statement 3 are correct

Mean derivation is least when measured about mean, therefore statement 3 is wrong.

Sol. 120. (d)

As standard deviation is '0', therefore all observations will be equal to 24.

Sol. 121. (a)

We know that the value of regression coefficient of y on x and the value of regression coefficient of x on y have same signs. Thus, (1,1) is the appropriate pair.

Sol. 122. (c)

$$\bar{x} = 10, \bar{y} = 90$$

$$\sigma_x = 3, \sigma_y = 12$$

$$r_{xy} = 0.8$$

Regression equation x on y is:

$$\Rightarrow x - 10 = r \frac{\sigma_x}{\sigma_y} (y - 90)$$

$$\Rightarrow x - 10 = 0.8 \times \frac{3}{12} (y - 90)$$

$$\Rightarrow x - 10 = 0.2 \times \frac{3}{12} (y - 90)$$

$$\Rightarrow x = -8 + 0.2y$$

Sol. 123. (b)

Total expenditure of A = 10,000

Total expenditure of B = 8,100

So, area of A : area of B = $10000:8100 = 100:81$

$\Rightarrow$ radii of A: radii of B = $\sqrt{100} : \sqrt{81} = 10:9$

Sol. 124. (d)

The arithmetic mean

$$= \frac{0 \times 1 + 1 \times {}^nC_1 + 2 \times {}^nC_2 + \dots + n \times {}^nC_n}{1 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n}$$

$$= \frac{n \cdot 2^{n-1}}{2^n} = \frac{n}{2}$$

Sol. 125. (c)

$$\frac{x_1 + x_2}{2} - \sqrt{x_1 x_2} > 1$$

$$\Rightarrow \frac{x_1 + x_2}{2} > \sqrt{x_1 x_2} + 1$$

$$\Rightarrow x_1 + x_2 > 2\sqrt{x_1 x_2} + 2$$

$$\Rightarrow x_1 + x_2 - 2\sqrt{x_1 x_2} > 2$$

$$\Rightarrow (\sqrt{x_1} - \sqrt{x_2})^2 > 2$$

$$\Rightarrow |\sqrt{x_1} - \sqrt{x_2}| > \sqrt{2}$$

Sol. 126. (b)

Variance is independent of change of origin but not scale. So, statement 1 is incorrect and statement 2 is correct.

Sol. 127. (c)

$$\text{Coefficient of correlation} = \sqrt{0.2 \times 1.8} = 0.6$$

Sol. 128. (b)

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^2$$

$$= \frac{1}{n} (1^2 + 2^2 + \dots + n^2) - \left(\frac{1}{n} (1 + 2 + 3 + \dots + n) \right)^2$$

$$= \frac{1}{n} \times \frac{n(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2 - 1}{12}$$

$$\Rightarrow \sigma = \sqrt{\frac{n^2 - 1}{12}}$$

Sol. 129. (d)

Average speed

$$= \frac{5 + 15}{\frac{5}{30} + \frac{15}{45}} = \frac{20 \times 90}{15 + 30} = 40 \text{ km/hr}$$

Sol. 130. (b)

New mean = $5 + 5 = 10$

New σ = Old $\sigma = 2$

$$\text{Coefficient of variation} = \frac{\sigma}{\text{Mean}} \times 100$$

$$= \frac{2}{10} \times 100 = 20$$

Sol. 131. (c)

20% of $360^\circ = 72^\circ$

Sol. 132. (b)

On arranging these 12 numbers in ascending order, the sixth and seventh terms are 4.6 and 5.2.

$$\therefore \text{Median} = \frac{4.6 + 5.2}{2} = 4.9$$

Sol. 133. (c)

By the properties of deviation

Sol. 134. (d)

Mean changes with changes in origin

Variance is independent to the choice of origin.

Sol. 135. (d)

Mean of $x_i = \bar{x}$

Mean of $a + cy_i = \bar{x}$

Mean of $cy_i = \bar{x} - a$

$$\text{Mean of } y_i = \frac{\bar{x} - a}{c}$$

Sol. 136. (d)

Sol. 137. (c)

by solving both linear equation $x = 13$ and $y = 17$

Sol. 138. (d)

$$r = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x)} \sqrt{\text{var}(y)}}$$

$$0.6 = \frac{27}{\sigma(x) \cdot 5}$$

$$\sigma(x) = 9$$

$$\text{var}(x) = 81$$

Sol. 139. (c)

by options

if we take numbers 9, 1 then their AM is 5 and GM is 3

Sol. 140. (d)

$x = \text{M.D.}, y = \text{S.D.}$

$$\text{M.D.} = \frac{4}{5} \text{ S.D.} \Rightarrow x < y$$

Sol. 141. (b)

Median can be obtained from ogive.

Sol. 142. (d)

Combined Average

$$= \frac{500 \times 1860 + 600 \times 1750}{1100}$$

Combined Variance

$$\sigma = \sqrt{\frac{n_1(\sigma_{100}^2 + d_1^2) + n_2(\sigma_{150}^2 + d_2^2)}{n_1 + n_2}} \quad \dots(i)$$

$$\sigma^2 = \frac{500(81 + 3600) + 600(100 + 2500)}{1100} = \frac{(5 \times 3681) + (6 \times 2600)}{1100} \approx 3092$$

Sol. 143. (a)

$$(x_1, x_2, \dots, x_n)^{1/n} = P$$

$$(y_1, y_2, \dots, y_n)^{1/n} = Q$$

$$\frac{(x_1, x_2, \dots, x_n)^{1/n}}{(y_1, y_2, \dots, y_n)^{1/n}} = \frac{P}{Q}$$

Sol. 144. (b)

Variance will not change by adding or subtracting a fixed value to all the elements.

Sol. 145. (c)

Sol. 146. (d)

Let No. of Men = M

Let No. of Women = W

$26M + 21W = 25(M + W)$

$M = 4W$

$M:W = 4:1$

Sol. 147. (d)

$r^2 = 0.64$

variation explained = 64%

not explained = 36%

Sol. 148. (b)

$$r = \sqrt{b_{xy} b_{yx}}$$

$$-\frac{1}{2} = \sqrt{b_{xy}(-6)} \Rightarrow b_{xy} = -\frac{1}{24}$$

Sol. 149. (d)

for straight line correlation coefficients are +1 or -1.

Sol. 150. (c)

Sol. 151. (c)

profit of companies is inversely proportion to claim they provided.

Sol. 152. (c)

$r = 0$ when X and Y are independent

Sol. 153. (c)

mean depends on both origin and scale but S.D. is independent from origin.

Sol. 154. (a)

A.M. = median = mode for sym. distribution.

Sol. 155. (d)

Sol. 156. (c)

$$\text{variance}(V) = \frac{n^2 - 1}{12} = \frac{399}{12} = 33.25$$

after multiply with 3 variance = $9 \times 33.25 = 299.25$

Sol. 157. (b)

$$\text{median} = \frac{(x+1) + (x+3)}{2} = 42$$

$x = 40$

5th term = 41, 6th term = 43

Sol. 158. (a)

$$S.D. = \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N}\right)^2}$$

$$= \sqrt{\frac{5000}{10} - \left(\frac{100}{10}\right)^2} = 20$$

Sol. 159. (c)

$6x + y = 30$

$3x + 2y = 25$

by solving these two equations

$x = 35/9, y = 20/3$

this intersection point of equations is value of mean of series.

$\bar{x} = 35/9, \bar{y} = 20/3$

$b_{xy} = -1/6, b_{yx} = -3/2$

$$r = \sqrt{b_{xy} b_{yx}}$$

$$r = \sqrt{\left(-\frac{1}{6}\right)\left(-\frac{3}{2}\right)} = -0.5$$

Sol. 160. (c)

limits are $m \pm \frac{d}{2}$

here $d = 5$

interval of first class is $5 \pm \frac{5}{2} = 5 \pm 2.5$

third option is correct.

Sol. 161. (b)

mean = 4.4

then sum of observations = $4.4 \times 5 = 22$

sum of given observations = $1 + 2 + 6 + x + y = 22$

$x + y = 13$

$y = 13 - x$

$$\text{variance} = \left(\frac{\sum x^2}{N}\right) - \left(\frac{\sum x}{N}\right)^2$$

$$\left(\frac{1^2 + 2^2 + 6^2 + x^2 + (13-x)^2}{5}\right) - \left(\frac{22}{5}\right)^2 = 8.24$$

by solving it $x = 4$ then $y = 9$

Sol. 162. (b)

x	f	cf
1	3	3
2	15	18
3	45	63
4	57	120
5	50	170
6	36	206
7	25	231
8	9	240

$N = 240/2 = 120$

cf just greater than 120 is 170 and corresponding value of x is 5. so median = 5

Sol. 163. (c)

S.D. is independent from change of origin.

Sol. 164. (c)

Range is independent from change of origin but depends on change of scale.

Sol. 165. (c)

$$\text{mean}(M) = \frac{n+1}{2} = \frac{15+1}{2} = 8$$

$$\text{variance}(V) = \frac{n^2 - 1}{12} = \frac{224}{12} = \frac{56}{3}$$

$V + M^2 = 248/3$

Sol. 166. (b)

distance is same so

$$V_{av} = \frac{2uv}{u+v} = \frac{2(3v)(2v)}{3v+2v} = 2.4v \text{ km/hr}$$

Sol. 167. (b)

$$\text{Combined mean} = \frac{N_B \bar{x}_B + N_G \bar{x}_G}{N_B + N_G}$$

let number of boys are x then girls $150 - x$

$$60 = \frac{x \cdot 70 + (150 - x) \cdot 55}{150}$$

$x = 50$ then girls = 100

Sol. 168. (c)

largest difference is $26 - 15 = 11$ in 2014-15

Sol. 169. (a)

maximum frequency in physics is 30 so modal class is 30-40

median of physics lie in 40-50 and median of mathematics lie in 30-40 so second statement is correct.

Sol. 170. (c)

Class	Phy.	x	fx
10-20	8	15	0120
20-30	11	25	0275
30-40	30	35	1050
40-50	26	45	1170

50-60	15	55	0825
60-70	10	65	0650
			sum = 4090

mean = $4090/100 = 40.9$

Sol. 171. (b)

$$\sigma = \sqrt{\frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2}$$

$$\sigma = \sqrt{\frac{32}{8} - \left(\frac{0}{8}\right)^2} = 2$$

Sol. 172. (d)

$$\sigma = \sqrt{\frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2}$$

$$\sigma = \sqrt{\frac{200}{10} - \left(\frac{20}{10}\right)^2} = 4$$

coeff. of variation

$$= \frac{\sigma}{\bar{x}} = \frac{4}{2} \times 100 = 200$$

Sol. 173. (c)

sum of 100 observations = 4000

after correction sum will be $4000 - 83 + 43 = 3960$

correct mean = $3960/100 = 39.6$

Sol. 174. (b)

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = 0.8 \times \frac{3.5}{2.5} = 1.12$$

regression line $y - 67 = 1.12(x - 65)$

$y = 1.12x - 5.8$

Sol. 175. (b)

by combined mean formula

$$\frac{4(x) + 9(x-1)}{x + (x-1)} = 6$$

$$4x + 9x - 9 = 12x - 6$$

$x = 3$

Sol. 176. (d)

$$\text{mean} = 2.5 + \frac{50}{n} = 3.5 + \frac{-50}{n}, n = 100$$

Sol. 177. (d)

$$\frac{n(2M) + 2n(M)}{n + 2n} = \frac{4Mn}{3n} = \frac{4}{3}M$$

Sol. 178. (b)

mean will increase by k

Sol. 179. (c)

algebraic sum of deviations from mean is always zero.

Sol. 180. (d)

$5x = 4y$

$$5 \frac{2mn}{m+n} = 4\sqrt{mn}$$

$$100m^2n^2 = 16mn(m+n)^2$$

$$25mn = 4(m^2 + n^2 + 2mn)$$

$$4m^2 - 17mn + 4n^2 = 0$$

$$4m^2 - 16mn - mn + 4n^2 = 0$$

$$4m(m-4n) - n(m-4n) = 0$$

$$(m-4n)(4m-n) = 0$$

$$m = 4n, 4m = n$$

$$m = 4n, 4m = n$$

Sol. 181. (a)

$$\text{coeff. of variation} = \frac{\sigma}{\bar{x}} = \frac{\sigma}{100} \times 100 = 45$$

$$\sigma = 45, \text{variance} = (45)^2 = 2025$$

Sol. 182. (b)

check 2nd options 6, 18, 18, 18, 30

mean, mode, median of above data is 18

Sol. 183. (b)

by combined mean formula $\frac{12(75) - 2(\bar{x})}{12 - 2} = 65$

$$\bar{x} = 125$$

Sol. 184. (c)

$$\text{new GM} = 3(10)^4 = 30000$$

Sol. 185. (c)

when 8 will be replaced by 3 then sum of observation will decrease by 50
so avg will decrease by $50/15 = 10/3$

Sol. 186. (d)

Sol. 187. (c)

Sol. 188. (b)

$$\text{Science graduates} = \frac{30}{150} \times 360^\circ = 72^\circ$$

Sol. 189. (a)

Sol. 190. (d)

Sol. 191. (a)

peas	f	cf
1	4	4
2	33	37
3	76	113
4	50	163
5	26	189
6	8	197
7	1	198

$$N = 198, N/2 = 99$$

cf more than 99 is 113

so median is 3

Sol. 192. (c)

Standard deviation

Sol. 193. (c)

Given regression lines are

$$3x - 4y + 8 = 0 \quad \dots(1)$$

$$\text{and } 4x - 3y - 4 = 0 \quad \dots(2)$$

When we rearrange

$$(1) \text{ will give } 4y = 3x + 8$$

$$y = \frac{3x + 8}{4}$$

$$\text{and } (2) \text{ will give } 4x = 4 + 3y$$

$$x = 1 + \frac{3}{4}y$$

so both one and 2 are correct.

Sol. 194. (a)

$$\text{Here } b_{xy} = 3/4$$

$$b_{yx} = 3/4$$

$$r = \sqrt{b_{xy} \cdot b_{yx}} = 3/4$$

Only 1 is correct.

Sol. 195. (c)

Marks	No. of students	C.f.
15-20	4	4
20-25	5	9
25-30	11	20
30-35	6	26
35-40	5	31
40-45	8	39
45-50	9	48
50-55	6	54
55-60	4	58
60-65	2	60
	N=60	

$$\frac{N}{2} = \frac{60}{2} = 30$$

$$\text{Median} = \ell + \left(\frac{\frac{N}{2} - C.f}{f_m} \right) \times h$$

$$= 35 + \frac{(30 - 26) \times 5}{5}$$

$$= 35 + 4$$

$$= 39$$

Most frequently occurring value is 11 of frequency.

Sol. 196. (b)

$$\text{Mode} = \ell + \left(\frac{f_m - f_o}{2f_m - f_o - f_1} \right) \times h$$

$$= 25 + \left(\frac{11 - 5}{22 - 5 - 6} \right) \times 6$$

$$= 25 + \frac{6}{11} \times 5$$

$$= 25 + \frac{30}{11}$$

$$= 25 + 2.72$$

$$= 27.73$$

Sol. 197. (i)

Mean of natural no's is [15, 64]

15, 16, 17, 64

$$\text{Mean} = \frac{\text{sum of all observations}}{\text{total no. of observations}}$$

$$= \frac{15 + \dots + 64}{60}$$

$$= \frac{60}{2} [15 + 64]$$

$$= \frac{60}{2} [79]$$

$$= \frac{60}{2} [79]$$

$$\text{As } 15 + 16 + \dots + 64$$

From A.P.

$$\text{So } S_n = \frac{n}{2} [a + \ell] =$$

$$\frac{30[15 + 64]}{60} = \frac{79}{2} = 39.5$$

Sol. 198. (a)

Let us consider x as 1

x, x, x + 2, x + 3, x + 10

so given terms become

1, 1, 3, 4, 11

Mean

$$= \frac{\text{sum of all obs.}}{\text{total no. of obs.}} = \frac{1 + 1 + 3 + 4 + 11}{5} = 4$$

Median: as the no. of obs. are odd

$$\text{So Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ obs.}$$

Mpde: most frequency occurring value is 1

Here statement (i) Mean > Mode is $4 > 1$

So it true.

2. Median > Mean is false

As median comes out to be 3 which is less than 4

Sol. 199. (b)

$$\bar{x} = 5.5 \text{ (Mean) } n = 10$$

If each obs. is multiplied by 4

then mean becomes $\bar{x} \times 4 = 22$

after subtracting from 44

$$= 44 - 22 = 22$$

Sol. 200. (c)

G.M. between 2 no. is $(ab)^{1/2}$

between n no's is $(x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n}$

$$\text{So here G.M. } (2^1 \cdot 2^2 \cdot 2^3 \cdot 2^4 \cdot 2^5 \cdot 2^6 \cdot 2^7 \cdot 2^8 \cdot 2^9 \cdot 2^{10})^{1/10}$$

$$= (2^{1+2+\dots+10})^{1/10}$$

$$= 2 \frac{55}{10}$$

$$= 2^5 \sqrt{2}$$

$$32 \sqrt{2}$$

Here $1 + 2 + \dots + 10$

Is sum 10 natural

$$\text{i.e. } \frac{10(10+1)}{2}$$

$$5 \times 11 = 55$$

So which greater than 32

$$32 < 55 < 64$$

Sol. 201. (c)

$$\text{H.M. between 2 no's is } a \text{ \& } b \text{ is } \frac{2ab}{a+b}$$

$$a = 60, b = x$$

$$\text{H.M.} = \frac{2(60)(x)}{60+x}$$

$$48 = \frac{120x}{60+x}$$

$$48(60+x) = 120x$$

$$288 + 48x = 120x$$

$$288 = 120x - 48x$$

$$72x = 2880$$

$$x = \frac{2880}{72}$$

$$x = 40$$

Sol. 202. (a)

first term even natural no's are

2, 4, 6, 8, 10, 12, 14, 16, 18, 20

$$N = 10$$

$$\bar{x} = \frac{2 + 40 + \dots + 20}{10}$$

$$= \frac{10}{2} [2 + 20] = 11$$

$$= \frac{10}{2} [2 + 20] = 11$$

Mean deviation

$$= \frac{\sum |x_i - \bar{x}|}{n} = \frac{2(9 + 7 + 5 + 3 + 1)}{10} = \frac{25}{5} = 5$$

Sol. 203. (b)

$$\text{Variance} = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$= \frac{1540}{10} - \left(\frac{110}{10} \right)^2 = 154 - 121 = 33$$

Sol. 204. (a)



Positive skew curve, Mean > median > mode

Sol. 205. (c)

Variance of 2, 2, 2, 5, x is 3.6

$$\text{Variance} = \frac{\sum x^2}{N} - \left(\frac{\sum x}{N} \right)^2$$

$$3.6 = \frac{4 + 4 + 1.6 + 25 + x^2}{5} - \left(\frac{13 + x}{5} \right)^2$$

$$10 = 5(49 + x^2) - (13 + x)^2$$

$$10 = 245 + 5x^2 - 169 - x^2 - 26x$$

$$4x^2 - 26x - 14 = 0$$

$$2x^2 - 13x - 7 = 0$$

$$2x^2 - 14x + x - 7 = 0$$

$$2x(x-7) + 1(x-7) = 0$$

$$x=7 \text{ is given in options.}$$

Sol. 206. (d)

$$\text{Arithmetic mean of A.P.} = \frac{a + \ell}{2}$$

$$= \frac{4 + (4 + (50-1)4)}{2}$$

$$= \frac{4+4+200-4}{2} = 102$$

Sol. 207. (b)

x	x - $\bar{x}$
20	10
21	9
23	7
26	4
27	3
33	3
34	4
37	7
39	9
40	10

$$\sum |x - \bar{x}| = 66$$

$$\bar{x} = \frac{\sum x}{N} = \frac{300}{10} = 30$$

$$\text{Mean deviation} = \frac{66}{10} = 6.6$$

$$\text{Coefficient of mean deviation} = \frac{\text{mean deviation}}{\text{mean}} = \frac{6.6}{30} = 0.22$$

Sol. 208. (d)

$$\text{Mean} = A + \frac{\sum d}{N}$$

$$\bar{x} = 100 + \frac{-20}{N} = 92 + \frac{140}{N}$$

$$\Rightarrow 8 = \frac{160}{N}$$

$$\Rightarrow N = 20$$

$$\bar{x} = 100 + \frac{-20}{20} = 99$$

Sol. 209. (a)

$$\bar{x} = A + \frac{\sum d}{N}$$

$$99 = 99 + \frac{\sum d}{N} \Rightarrow \sum d = 0$$

Sol. 210. (c)

$$99 = y + \frac{180}{20}$$

$$99 = y + 9, y = 90$$

Sol. 211. (d)

$$4, 7, 10, 13, 16, \dots \text{Upto 51 terms}$$

$$\text{Mean} = \frac{A + \ell}{2}$$

$$= \frac{4 + [4 + (51-1)3]}{2} = 79$$

Sol. 212. (b)

$$\text{Median is } \frac{51+1}{2} = 26\text{th term}$$

$$T_{26} = 4 + (26-1)3 = 4 + 75 = 79$$

Sol. 213. (b)

Mean = median

Sum of deviations from mean is always zero

Sol. 214. (c)

$$\sum f = 120$$

$$17 + p + q + 32 + p - 3q + 19 = 120$$

$$2p - 2q + 68 = 120$$

$$2p - 2q = 52$$

$$p - q = 26 \quad \dots (1)$$

Class	x	f	fx
0-20	10	17	170
20-40	30	p + q	30p + 30q
40-60	50	32	1600

60-80	70	p - 3q	70p - 210q
80-100	90	19	3480 + 100p - 180q

$$\bar{x} = 50 = \frac{3480 + 100p - 180q}{120}$$

$$\Rightarrow 6000 = 3480 + 100p - 180q$$

$$\Rightarrow 5p - 5q = 126 \quad \dots (2)$$

By solving (1) & (2)

$$q = 1 \text{ \& } p = 27$$

Sol. 215. (a)

Solved in above question

Sol. 216. (b)

No effect on mean by doubled the frequency

Sol. 217.

mean of square of first n natural numbers is

$$\frac{(n+1)(2n+1)}{6} = x$$

$$\text{mean of first n natural numbers is } \frac{(n+1)}{2} = y$$

$$\frac{x}{y} = \frac{\frac{(n+1)(2n+1)}{6}}{\frac{(n+1)}{2}} = \frac{2(2n+1)}{3(n+1)} = \frac{55}{42}$$

by solving above linear equation n = 27

Sol. 218. (c)

coefficient of correlation is independent from change of origin

so its will not affected by adding 25 in each terms i.e. r will be remains 0.7

Sol. 219. (d)

$$\text{variability depends upon } \frac{\sigma}{\bar{x}} \times 100$$

coefficient of variation

$$\text{for maths} = \frac{15}{40} \times 100 = 37.5\%$$

$$\text{for physics} = \frac{12}{28} \times 100 = 42.85\%$$

$$\text{for chemistry} = \frac{14}{38} \times 100 = 36.84\%$$

$$\text{for biology} = \frac{16}{36} \times 100 = 44.44\%$$

maximum variability is in biology.

Sol. 220. (a)

coefficient of variation

$$\text{for maths} = \frac{15}{40} \times 100 = 37.5\%$$

Sol. 221. (c)

Class	x	f	cf
0-10	5	1	1
10-20	15	2	3
20-30	25	4	7
30-40	35	6	13
40-50	45	4	17
50-60	55	3	20
		$\sum f = 20$	

median class is 30-40

$$m = 30 + \frac{\frac{20}{2} - 7}{6} \times 10 = 35$$

Sol. 222. (d)

x	f	x - m	f x - m
5	1	30	30
15	2	20	40
25	4	10	40
35	6	0	0
45	4	10	40

55	3	20	60
	$\sum f = 20$		210

M.D. about median = $210/20 = 10.5$

Sol. 223. (b)

x	f	x - $\bar{x}$	f x - $\bar{x}$
5	1	29.5	29.5
15	2	19.5	39
25	4	9.5	38
35	6	0.5	3
45	4	10.5	42
55	3	20.5	61.5
	$\sum f = 20$		213

mean deviation about mean = $213/20 = 10.65$

Sol. 224. (b)

x	f	cf
1	4	4
2	6	10
3	9	19
5	7	26
	$N = 26$	

cf greater than N/2 is 10, that is corresponding to x = 2, so median = 2

Sol. 225. (d)

median of -1, 1, 4, 3, 8 is 3

because median is middle value when data is placed in ascending order.

so M = 3

median of 12, 17, 19, 9, 11 is 12

so N = 12

so 4M - N = 12 - 12 = 0

Sol. 226. (d)

$$\text{let } 5P = 4Q = \frac{R}{2} = k$$

$$P = k/5, Q = k/4, R = 2k$$

$$\frac{P+Q}{2P+0.7R} = \frac{\frac{k}{5} + \frac{k}{4}}{2\left(\frac{k}{5}\right) + 0.7(2k)} = \frac{\frac{9}{20}}{\frac{18}{10}} = \frac{1}{4}$$

Sol. 227. (d)

$$G = (1.2.2^2.2^3.2^4 \dots 2^{n-1})^{\frac{1}{n}}$$

$$G = (2^{0+1+2+3+4+\dots+n-1})^{\frac{1}{n}}$$

$$G = \left(2^{\frac{n(n-1)}{2}}\right)^{\frac{1}{n}} = (2)^{\frac{n-1}{2}}$$

$$1 + 2\log_2 G = 1 + 2\log_2 (2)^{\frac{n-1}{2}} = 1 + 2\left(\frac{n-1}{2}\right) = n$$

Sol. 228. (b)

$$H = \frac{n}{\frac{1}{1} + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{n-1}}}$$

$$\frac{n}{H} = \frac{1}{1} + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{n-1}}$$

above series is a GP, so by formula of sum of GP

$$\frac{n}{H} = \frac{1\left(1 - \frac{1}{2^n}\right)}{1 - \frac{1}{2}} = 2\left(1 - \frac{1}{2^n}\right) = 2 - \frac{1}{2^{n-1}}$$

Sol. 229. (c)

$$S = \sum_{i=1}^n (2x_i - a)^2 = (2x_1 - a)^2 + (2x_2 - a)^2 + (2x_3 - a)^2 + (2x_4 - a)^2 + \dots + (2x_n - a)^2$$

for minimum or maximum value differentiation should be zero.

$$\frac{dS}{dx} = 4(2x_1 - a) + 4(2x_2 - a) + 4(2x_3 - a) + 4(2x_4 - a) + \dots + 4(2x_n - a) = 0$$

$$= 8(x_1 + x_2 + x_3 + \dots + x_n) - 4(na) = 0$$

$$= 8(nQ) - 4(na) = 0$$

$$= a = 2Q$$

Sol. 230. (c)

mean of 5 observations = 14

so sum of five observations = 70

five observations are 11, 16, 20, x, 23 - x

$$\text{variance} = \frac{\sum x^2}{n} - (\text{mean})^2$$

$$13.2 = \frac{121 + 256 + 400 + x^2 + (23 - x)^2}{5} - 196$$

$$66 = 121 + 256 + 400 + x^2 + (23 - x)^2 - 980$$

$$x^2 + (23 - x)^2 = 269$$

by solving above quadratic equation

x = 10, 13

Sol. 231. (c)

y is given and we have to find x

i.e. regression line x on y is required.

let $x + 4y + 1 = 0$ is line x on y

$$b_{xy} = -4$$

and $4x + 9y + 7 = 0$ is line y on x

$$b_{yx} = -4/9$$

$b_{xy} \cdot b_{yx} > 1$ i.e. we assumed incorrectly.

so $4x + 9y + 7 = 0$ is regression line x on y, so

put $y = -3$ in this equation then $x = 5$.

Sol. 232. (d)

let $9p = 3q = 2r = 6s = k$

we know that

$$p + q + r + s = 360^\circ$$

$$\frac{k}{9} + \frac{k}{3} + \frac{k}{2} + \frac{k}{6} = 360^\circ$$

$$\frac{20k}{18} = \frac{10k}{9} = 360^\circ$$

$$4p - q = \frac{4k}{9} - \frac{k}{3} = \frac{k}{9} = 36^\circ$$

Sol. 233. (a)

data in ascending order

1, 1, 1, 2, 3, 3, 4, 4, 4, 5, 6, 6

$m = 19/8 = 2.375$ and $M = 21/4 = 5.25$

$2m + M = 4.75 + 5.25 = 10$

Sol. 234. (a)

line of regression y on x

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y - 1 = \frac{2 - 1}{3 + 1} (x + 1)$$

$$4y - 4 = x + 1$$

$$x - 4y + 5 = 0$$

Sol. 235. (d)

mode of 1, 2, 1, 1, 2, 1, 4, 6, 5, 4 is 1.

Sol. 236. (a)

$$(3 + 2\sqrt{2})x^2 - (4 + 2\sqrt{3})x + (8 + 4\sqrt{3}) = 0$$

let α and β are roots of equation

$$\text{we know that } \alpha + \beta = \frac{-b}{a} = \frac{(4 + 2\sqrt{3})}{(3 + 2\sqrt{2})}$$

$$\text{and } \alpha\beta = \frac{c}{a} = \frac{(8 + 4\sqrt{3})}{(3 + 2\sqrt{2})}$$

$$H.M. = \frac{2\alpha\beta}{\alpha + \beta} = \frac{c/a}{-b/a}$$

$$= -\frac{c}{b} = \frac{(8 + 4\sqrt{3})}{(4 + 2\sqrt{3})} = 2$$

Sol. 237. (a)

GM of roots or equation

$$= \sqrt{\alpha\beta} = \sqrt{\frac{(8 + 4\sqrt{3})}{(3 + 2\sqrt{2})}}$$

$$\Rightarrow \sqrt{\frac{2(4 + 2\sqrt{3})}{(3 + 2\sqrt{2})} \times \frac{(3 - 2\sqrt{2})}{(3 - 2\sqrt{2})}}$$

$$\Rightarrow \sqrt{\frac{2(\sqrt{3} + 1)^2}{9 - 8} \times \frac{(\sqrt{2} - 1)^2}{1}} = \sqrt{2}(\sqrt{3} + 1)(\sqrt{2} - 1)$$

$$= 2(\sqrt{6} - \sqrt{3} + \sqrt{2} - 1)$$

Sol. 238. (a)

mean by indirect method

$$\bar{x} = A + \frac{\sum d}{n}$$

deviation by 10 is p

$$\bar{x} = 10 + \frac{p}{n} \quad \dots\dots\dots(i)$$

deviation by 20 is q

$$\bar{x} = 20 + \frac{q}{n} \quad \dots\dots\dots(ii)$$

equate both above equations

$$10 + \frac{p}{n} = 20 + \frac{q}{n}$$

$$\frac{p - q}{n} = 10$$

$(p - q)^2 = 10000$ given so $p - q = 100$

$$\frac{100}{n} = 10$$

$$n = 10$$

Sol. 239. (c)

$$\sum_{i=1}^{10} \left(\frac{3x_i - 4}{5} \right) = \left(\frac{3x_1 + 4}{5} \right) + \left(\frac{3x_2 + 4}{5} \right)$$

$$+ \left(\frac{3x_3 + 4}{5} \right) + \dots + \left(\frac{3x_{10} + 4}{5} \right)$$

$$= \frac{3}{5} (x_1 + x_2 + \dots + x_{10}) + \left(\frac{4}{5} + \frac{4}{5} + \frac{4}{5} + \dots + \frac{4}{5} \right)$$

$$= \frac{3}{5} (200) + \left(\frac{40}{5} \right) = 112$$

Sol. 240. (d)

$$\sigma = \sqrt{\frac{\sum x^2}{n} - (\text{mean})^2}$$

$$\sigma = \sqrt{\frac{16160}{10} - (40)^2} = \sqrt{16} = 4$$

Sol. 241. (c)

x	f	fx
1	${}^9C_0 = 1$	1
2	${}^9C_1 = 9$	18
3	${}^9C_2 = 36$	108
4	${}^9C_3 = 84$	336
5	${}^9C_4 = 126$	630
6	${}^9C_5 = 126$	756
7	${}^9C_6 = 84$	588
8	${}^9C_7 = 36$	288
9	${}^9C_8 = 9$	81
10	${}^9C_9 = 1$	10

	512	2816
--	-----	------

$$\text{mean} = 2816/512 = 5.5$$

Sol. 242. (a)

coefficient of correlation is independent from change of origin and change of scale, but depends upon change of sign.

sign of y observation is changing from y to -y so now coefficient of correlation will be -r.

Sol. 243. (c)

$$H = \frac{3}{\frac{1}{{}^{10}C_4} + \frac{1}{{}^{10}C_5} + \frac{1}{{}^{10}C_6}}$$

$$\frac{3}{H} = \frac{1}{{}^{10}C_4} + \frac{1}{{}^{10}C_5} + \frac{1}{{}^{10}C_6}$$

we know that ${}^{10}C_4 = {}^{10}C_6$

$$\frac{3}{H} = 2 \frac{4 \times 3 \times 2 \times 1}{10 \times 9 \times 8 \times 7} + \frac{5 \times 4 \times 3 \times 2 \times 1}{10 \times 9 \times 8 \times 7 \times 6}$$

$$\frac{270}{H} = 2 \frac{3}{7} + \frac{5}{7 \times 2} = \frac{17}{4}$$

Sol. 244. (b)

data (-4, 1), (-1, 2), (2, 7) and (3, 1)

regression line y on x

$$y - y_1 = b_{yx} (x - x_1)$$

here x_1 is mean of data x that is 0

y_1 is the mean of data y that is 11/4

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = \frac{\sum XY}{N \sigma_x \sigma_y} \cdot \frac{\sigma_y}{\sigma_x} = \frac{\sum XY}{N \sigma_x^2}$$

x	y	$X = x - x_1$	$Y = y - y_1$	XY
-4	1	-4	-7/4	28/4
-1	2	-1	-3/4	3/4
2	7	2	17/4	34/4
3	1	3	-7/4	-21/4
				11

$$b_{yx} = \frac{\sum XY}{N \sigma_x^2} = \frac{11}{4 \times \frac{30}{4}} = \frac{11}{30}$$

$$y - \frac{11}{4} = \frac{11}{30} (x)$$

$$y = \frac{11}{4} + \frac{11}{30} (x)$$

a = 11/4 and b = 11/30

so $2a + 15b = 11$

Sol. 245. (a)

angle between two straight line by the concept of 2D geometry

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{-\frac{1}{2} + \frac{2}{3}}{1 + \frac{1}{2} \cdot \frac{2}{3}} = \frac{1}{8}$$

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = \frac{3 \frac{1}{8} - \frac{1}{64}}{1 - 3 \frac{1}{64}} = \frac{191}{488}$$

$$488 \tan 3\theta = 191$$

Sol. 246. (a)

let the equal amount is x

oil sold at the rate 150 is x/150

oil sold at the rate 200 is x/200

oil sold at the rate 250 is x/250

oil sold at the rate 300 is x/300

avg price of oil

= total amount / total quantity

$$= \frac{x+x+x+x}{\frac{x}{150} + \frac{x}{200} + \frac{x}{250} + \frac{x}{300}} = 210.52$$

Sol. 247. (b)

let $n = 3$

x_i	1	2	3
f_i	1	2^{-1}	2^{-2}
$x_i f_i$	1	$\frac{2}{2}$	$\frac{3}{4}$

$$S = 1 + \frac{2}{2} + \frac{3}{4} = \frac{11}{4}$$

now by checking options

put $n = 3$ in 1st option and its satisfied.

Sol. 248. (c)

$$\text{mean} = \frac{\sum fx}{\sum f} = \frac{11/4}{7/4} = \frac{11}{7}$$

now by checking options

put $n = 3$ in 3rd option and its satisfied.

Sol. 249. (c)

largest 5 observations are 32, 35, 41, 47, 50

x	$ x - \bar{x} $	$ x - \bar{x} ^2$
32	9	81
35	6	36
41	0	0
47	6	36
50	9	81
	$\sum (x - \bar{x}) = 30$	$\sum (x - \bar{x})^2 = 234$

$$\text{mean} = \frac{\sum x}{N} = \frac{205}{5} = 41$$

$$\text{mean deviation} = \frac{\sum (x - \bar{x})}{N} = \frac{30}{5} = 6$$

Sol. 250. (d)

$$\text{variance} = \frac{\sum (x - \bar{x})^2}{N} = \frac{234}{5} = 46.8$$

Sol. 251. (b)

$$\sigma = \sqrt{\frac{\sum x^2}{n} - (M)^2}$$

$$\sigma^2 + M^2 = \frac{\sum x^2}{n} = \frac{855}{9} = 95$$

Sol. 252. (d)

given that mean of n data is $\bar{x}$

so sum of data will be $n\bar{x}$

x_n is replace by k so new sum is $n\bar{x} - x_n + k$

$$\text{new mean} = \frac{n\bar{x} - x_n + k}{n}$$

Sol. 253. (d)

$$x - 3y + 4 = 0 \text{ and } 2x - 7y + 8 = 0$$

let $x - 3y + 4 = 0$ is line x on y

$$b_{xy} = 3$$

and $2x - 7y + 8 = 0$ is line y on x

$$b_{yx} = 2/7$$

$b_{xy} \cdot b_{yx} < 1$ i.e. we assumed correctly.

$$b_{xy} + 7b_{yx} = 3 + 2 = 5$$

Sol. 254. (c)

$$\bar{x} = \frac{1+4+9+16+\dots+n^2}{n}$$

$$\bar{x} = \frac{n(n+1)(2n+1)}{6n} = \frac{(n+1)(2n+1)}{6}$$

$$130 = \frac{(n+1)(2n+1)}{6}$$

by solving above quadratic equation
 $n = 20$

Sol. 255. (b)

x	$ x - \bar{x} $
1	4.5
2	3.5
3	2.5
4	1.5
5	0.5
6	0.5
7	1.5
8	2.5
9	3.5
10	4.5
	$\sum (x - \bar{x}) = 25$

$$\text{mean} = \frac{\sum x}{N} = \frac{55}{10} = 5.5$$

$$\text{mean deviation} = \frac{\sum (x - \bar{x})}{N} = \frac{25}{10} = 2.5$$

Sol. 256. (d)

height	f	cf
160-162	12	12
162-164	15	27
164-166	24	51
166-168	13	64
	$\sum f = 64$	

24 students have height between 164 and 166, we cannot say how many are there who have height between 164 and 165, so answer is cannot determine.

Sol. 257. (c)

$$N/2 = 64/2 = 32$$

Median class 164 - 166

$$\text{median} = l + \frac{\frac{N}{2} - c}{f} \times h = 164 + \frac{32 - 27}{24} \times 10$$

$$= 164 + 0.41 = 164.41$$

Sol. 258. (d)

Modal class 164 - 166

$$\text{median} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

$$= 164 + \frac{24 - 15}{48 - 15 - 13} \times 10 = 164 + 0.9 = 164.9$$

Sol. 259. (c)

Histogram is suitable for continuous frequency series.

Sol. 260. (b)

$$\text{var}(x) = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2$$

$$\text{var}(x) = \frac{900}{50} - \left(\frac{200}{50} \right)^2 = 18 - 16 = 2$$

$$\text{var}(y) = \frac{\sum y^2}{n} - \left(\frac{\sum y}{n} \right)^2$$

$$\text{var}(y) = \frac{1400}{50} - \left(\frac{250}{50} \right)^2 = 3$$

$\text{var}(x) < \text{var}(y)$

Sol. 261. (a)

Coeff. Of variation of X

$$= \frac{\sigma}{x} \times 100 = \frac{\sqrt{2}}{4} \times 100 = \frac{141.4}{4} = 35.6$$

Coeff. Of variation of Y

$$= \frac{\sigma}{y} \times 100 = \frac{\sqrt{3}}{5} \times 100 = \frac{173.2}{5} = 34.6$$

Coefficient of variation of X is strictly more than coefficient of variation of Y.

Sol. 262. (a)

Arithmetic mean is dependent on change of origin and scale, so new mean $(50 - 5)/20 = 2.25$.

Sol. 263. (b)

S.D. is only dependent on change of scale, so new S.D. $10/20 = 0.5$.

Sol. 264. (c)

$$\bar{x} = \frac{64+81+100+121+144+169+196+225}{8}$$

$$\bar{x} = \frac{1100}{8} = 137.5$$

Sol. 265. (a)

Harmonic mean

$$= \frac{5}{\frac{1}{10C_3} + \frac{1}{10C_4} + \frac{1}{10C_5} + \frac{1}{10C_6} + \frac{1}{10C_7}}$$

$$= \frac{5}{\frac{2}{10C_3} + \frac{2}{10C_4} + \frac{1}{10C_5}}$$

$$= \frac{5}{\frac{12}{10.9.8} + \frac{48}{10.9.8.7} + \frac{120}{10.9.8.7.6}} = \frac{3150}{19}$$

Sol. 266. (a)

Standard deviation is independent from change of origin.

Sol. 267. (b)

marks	x	f	fx
5 - 15	10	20	200
15 - 25	20	30	600
25 - 35	30	30	900
35 - 45	40	20	800
		$\sum f = 100$	$\sum fx = 2500$

$$\text{Mean} = 2500/100 = 25$$

Sol. 268. (b)

mean, median, mode, mean deviation, standard deviation have same unit of measurement but unit of variance is square of unit of measurement.